



Integration Guide | PUBLIC
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Integrating Joule with SAP Solutions

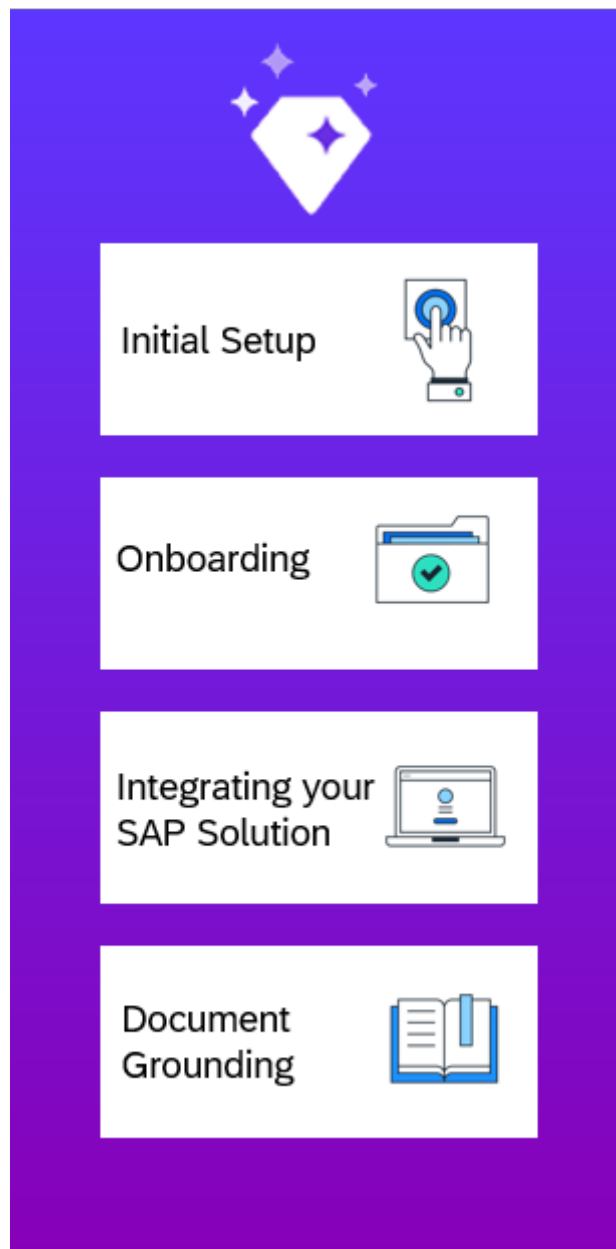
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1 Introduction

This guide provides detailed information to system administrators or technical consultants on how to set up Joule and the integration between Joule and other SAP products.



- [Check the setup requirements for your SAP Solution \[page 6\]](#)

- [Follow the Joule onboarding process to get Joule ready for integrating your SAP Solution \[page 7\]](#)
- [Integrate your SAP Solution with Joule to use the Joule capabilities your solution offers \[page 19\]](#)
- [Set up document grounding to upload documents to the Joule service \[page 70\]](#)

2 Initial Setup

Learn about the options for setting up Joule for productive purposes.

Before you perform the setup instructions, it's important to know which path you should take. Here are the different ways that Joule is available.

Joule is already available. I don't have to do anything.

You can tell that this applies to you when you see the Joule icon  inside your product. For example, in the SAP BTP cockpit, just click the Joule icon and start working with Joule.

I already have a BTP platform set up. I just need to run the Joule booster.

In this situation, you already have a global account, subaccounts, and entitlements set up. You also have users onboarded and trust set up to support your platform. Since the BTP platform is ready, you simply need to initiate the booster to prepare Joule and then configure the trusted domains.

Instructions:

- Make sure you [Configure Trust to the Identity Authentication Tenant \[page 10\]](#) and [Configure User Attributes for Joule from the Identity Directory \[page 11\]](#) before running the Joule booster.
- [Run the Joule Booster \[page 12\]](#)

I need to set up the BTP platform first and then I can onboard Joule.

If you haven't set up a BTP platform, then you'll need to set up BTP first. This involves administrative actions in the SAP BTP cockpit. After you set up the platform, you can onboard Joule.

Instructions:

- [Prerequisites \[page 7\]](#)
- [Joule Onboarding Guide \[page 7\]](#)

Related Information

[Joule Onboarding Guide \[page 7\]](#)

[Integration Guide \[page 4\]](#)

3 Joule Onboarding Guide

A step by step guide for setting up Joule and integrating it with SAP solutions.

Overview of the Onboarding Process

Joule includes multiple components that must be integrated for a seamless working solution. The onboarding process consists of the following steps:

[Prerequisites \[page 7\]](#)

Prerequisites for Joule Onboarding.

[Configure Trust to the Identity Authentication Tenant \[page 10\]](#)

Use your SAP Cloud Identity Services - Identity Authentication tenant as an Identity Provider or a proxy to your own Identity Provider hosting your business users.

[Configure User Attributes for Joule from the Identity Directory \[page 11\]](#)

A step by step guide to configure user attributes from the identity directory.

[Run the Joule Booster \[page 12\]](#)

Instructions for running the booster to set up Joule.

[Configure Trusted Domains in Identity Authentication \[page 15\]](#)

You need to add the domain of the integration product as trusted domain in the Identity Authentication.

[Configure Trusted Domain in SAP BTP \[page 16\]](#)

You can add, configure and save newly created trusted domain.

[Constraints for Joule \[page 16\]](#)

All constraints applicable to Joule are listed in this section. These constraints will provide you better understandability of Joule.

[Unsubscribe from Joule \[page 17\]](#)

This section provides instructions on how to unsubscribe and delete a Joule subscription and service instance. Use it to efficiently manage your subscriptions and free up resources when no longer needed.

3.1 Prerequisites

Prerequisites for Joule Onboarding.

Before completing the Joule onboarding tasks, you'll need to ensure you've met the required prerequisites for both your product and for Joule.

SAP Product Prerequisites

- You have a license for an SAP product that supports Joule integration (such as SAP SuccessFactors, SAP S/4HANA Cloud Private Edition. etc) in one of the supported data centers, as Joule is an embedded application that is integrated with SAP products. For more information, see [Data Centers Supported by Joule](#).

Note

- Joule is available with SAP Start under Early Adopter Care program only.
- If you want to integrate Joule with SAP Start, you must have a license of at least one SAP product that supports Joule integration.

- You have integrated the SAP product with Identity Authentication as Joule leverages the IAS setup of the SAP product for user login.
- You have an enterprise global account on SAP BTP as Joule is an SAP BTP based application and relies on SAP BTP services. For more information, see [Enterprise Accounts](#).

Note

If you already have a global account in SAP BTP, Joule entitlements can be added in the same global account.

Joule Specific Prerequisites

- You have the entitlements for Cloud Identity Services- Identity Authentication and Identity Provisioning. Joule uses the Identity Authentication for user login and Identity Provisioning service to handle provisioning of identities and their authorizations to various business applications (SAP SuccessFactors, Identity Authentication Service Tenant, SAP Build Work Zone, standard edition). For more information, see Tenant Models and licensing information for [Identity Authentication](#) and [Identity Provisioning](#).

Note

Please contact your account executive if any of the entitlements are not available in your global account.

- To support navigation to different applications from Joule and handle role based access scenarios, you need to configure the navigation service of SAP Build Work Zone. Verify that your global account is configured with the following entitlements. For more information, see [Managing Entitlements and Quotas Using the Cockpit](#).

Note

By default SAP Build Work Zone, standard edition uses the SAP Authentication and Trust Management service (XSUAA) for identity authentication. Therefore we recommend that you first switch to SAP Cloud Identity Services - Identity Authentication as the authentication mechanism. For more information, see [Switching to SAP Cloud Identity Services - Identity Authentication](#).

The Joule conversational search capability doesn't require integration with SAP Build Work Zone, standard edition.

Required Applications

Application	Technical Name	Plan	Required Quota	Remaining Quota
Joule	das-application	Standard	1	limited
SAP Build Work Zone, standard edition	SAPLaunchpad	Foundation or Standard	1	limited

Required Services

Service	Technical Name	Plan	Required Quota	Remaining Quota
SAP Build Work Zone, standard edition	build-workzone-standard	foundation or standard	1	limited

Note

All SAP BTP Global accounts already include entitlements for the foundation plan of SAP Build Work Zone, standard edition. Joule can only interact with one SAP Build Work Zone instance, therefore it is recommended to install Joule into an existing SAP Build Work Zone instance. If no SAP Build Work Zone instance exists you can start with foundation plan included in your SAP BTP Global account. Make sure that you choose the same plan for SAP Build Work Zone, standard edition for the application and service.

Joule uses the navigation service component of SAP Build Work Zone, standard edition to resolve intent based navigation targets and configure additional content providers. The content needs to be exposed by the SAP product (source system) and consumed by SAP Build Work Zone (target system).

- You have created a subaccount in your global account using the SAP BTP cockpit for provisioning Joule. A single Joule subscription can be integrated into multiple SAP products like SAP SuccessFactors, SAP S/4HANA Cloud Private Edition, or SAP Build Work Zone. However, one subscription of the Joule application can only be integrated with one tenant of an SAP product. This means you cannot include multiple systems of the same product in one Joule formation. You must ensure that there is one IAS tenant per single independent entity and operate every single independent entity from its own dedicated SAP BTP sub-account.

→ Tip

For more information, see [Create a Subaccount in the Cloud Foundry Environment](#).

Related Information

[SAP BTP Onboarding](#)

[SAP Cloud Identity Services](#)

[SAP Build Work Zone, standard edition](#)

[Joule Onboarding Guide \[page 7\]](#)

3.2 Configure Trust to the Identity Authentication Tenant

Use your SAP Cloud Identity Services - Identity Authentication tenant as an Identity Provider or a proxy to your own Identity Provider hosting your business users.

Prerequisites

- You have subaccount administrator permissions. For more information, see [Managing Subaccounts Using the Cockpit](#).
- You have a tenant of SAP Cloud Identity Services.
For more information, see [Tenant Model and Licensing](#) in the documentation for Identity Authentication.
- The Identity Authentication tenant is associated with the customer IDs of the relevant global account of SAP BTP.
For more information, see [Reuse SAP Cloud Identity Services Tenants for Different Customer IDs](#) in the documentation for Identity Authentication.
- You must use the same Identity Authentication tenant for all SAP applications that are included in the Joule formation.

Procedure

1. In the SAP BTP cockpit, go to your subaccount and choose ► [Security](#) ► [Trust Configuration](#) ►.
2. Choose [Establish Trust](#).
The [Configure Tenant](#) wizard opens.
3. Choose the Identity Authentication tenant. The Identity Providers listed are the Identity Authentication tenants associated with your customer ID. Click [Next](#).
4. Choose the domain configured with the Identity Authentication tenant and continue with [Next](#).

Note

For single-sign on to work, the domain of the Identity Authentication tenant must be same as the corresponding SAP system trust setup.

5. You can change the name and the description of the tenant, display and change the origin key, and provide a link text for user logon (see [Using Multiple Identity Providers from the Same Subaccount](#)). Click [Next](#).
6. Review your configuration and confirm using [Finish](#).

Related Information

[Establish Trust and Federation Between SAP Authorization and Trust Management Service and Identity Authentication](#)

3.3 Configure User Attributes for Joule from the Identity Directory

A step by step guide to configure user attributes from the identity directory.

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under [Applications and Resources](#), choose the [Applications](#) tile.
3. Select the application which was automatically created in the previous step.
4. If you don't have a created application in your list, you can create one. For more information, see [Create a New Application](#).
5. Choose the [Trust](#) tab.
6. Under [SINGLE SIGN-ON](#), choose [Attributes](#).
7. Under the [Self-defined Attributes](#) section, choose [Expand All](#) to view all the information about the user attributes. Ensure that the following SAML Assertion Attributes are present:

User Attribute	User Attribute Name
email	Email
Email_verified	Email Verified
family_name	Last Name
given_name	First Name
groups	Groups
user_uuid	Global User Id

8. If any attribute is missing, choose the [Add](#) button.
9. Provide a name for the attribute.
10. Choose [Identity Directory](#) source.
11. Choose a value from the drop-down list.
12. Optional: Choose the plus button next to the attribute to set a new value for the attribute.
13. Save your configuration.

Related Information

[SAP Cloud Identity Services - Configuring User Attributes from the Identity Directory](#)
[Global User ID in Integration Scenarios](#)

3.4 Run the Joule Booster

Instructions for running the booster to set up Joule.

Prerequisites

Before you run the booster, make sure:

- You have fulfilled all the necessary prerequisites for Joule. For more information, see [Prerequisites \[page 7\]](#).
- You have ensured that the SAP Systems where you want to integrate Joule, are registered in the *System Landscape* page in the SAP BTP cockpit. For more information, see [Maintaining Unified Customer Landscape](#).
- You must [Configure Trust to the Identity Authentication Tenant \[page 10\]](#) and [Configure User Attributes for Joule from the Identity Directory \[page 11\]](#).
- You have fulfilled the prerequisites for setting up integration with a selected product. Refer to the product specific integration section with which you want to integrate Joule.

Procedure

1. Log in to the SAP BTP account as global account administrator.
2. In the left navigation pane of the global account, click [Boosters](#).
3. In the Boosters screen, search for [Setting up Joule](#) and click [Start](#).
4. The following steps are involved in completing the provisioning process:

Steps	Action
Check Prerequisites	The following validations run in the backend: • Global account administrator role for running the booster • Service entitlements in the global account • Identity Authentication tenant to login to the integrated product
Set up Subaccount	Select the subaccount that should be used for Joule provisioning and configurations.

Steps	Action
Select Integrations	1. Select the SAP solutions where the Joule icon should be enabled. 2. Choose if this integration is used for testing or productive purposes. Note Providing the integration use case improves usage data analysis that is used to improve the Joule user experience. 3. Click Next.
Select Capabilities	1. Select the SAP solutions whose capability packages should be provisioned in Joule. 2. Choose if Joule should log the conversation data for your tenant. For more information, see Data Protection and Privacy. 3. Click Next.
Set Up Integrations	1. Provide a unique name for the Joule formation. 2. Select the systems for the SAP solutions that you had chosen in screens 3 and 4. Note • If you have selected SAP SuccessFactors or SAP S/4HANA Cloud Public Edition in the Select Capabilities screen, then you must select SAP Build Work Zone system to support the intent based navigation. For more information, see Create SAP Build Work Zone Application and Instance [page 20]. • You must enable the Identity Authentication for the Build Work Zone, Standard Edition. For more information, see Switching to SAP Cloud Identity Services - Identity Authentication. 3. Select the SAP Cloud Identity Services System that is configured in your SAP BTP Joule sub-account. 4. Click Next.
Validation	Some product specific validation run in the background. If the validation fails, you can either: • Go back to screen 4 and select a different system or, • Go back to screen 3 and select a different subaccount in which the Joule subscription should be created

Steps	Action
Review	Review the information that you have entered on the last page of the booster and click Finish to trigger Joule provisioning.

- After a successful provisioning, a confirmation message is displayed. A Joule formation is created in the [System Landscape](#) that includes the Joule system and the selected integration systems.

Include or Exclude a System in the Joule Formation

Note

This functionality is relevant only if the Joule Formation (Ready state) is available in your system landscape. If you have set up the integration with Joule, but the Joule Formation does not exist in the system landscape, you must run the Joule booster again.

If you have already set up integration with Joule, you can include or exclude systems in the existing Joule formation. For more information, see [Formations](#).

To include a system in an existing formation:

- Log in to the [SAP BTP account](#) as global account administrator.
- In the left navigation pane of the global account, click [System Landscape](#).
- Under System Landscape, click on [Formations](#).
- Choose an existing Joule formation (with formation type: Integration with Joule), which is in *Ready* state for which you wish to include an additional system. Click [Include Systems](#). The [Include Systems](#) dialog opens up.
 - Select the system which you want to include in the formation.
 - Click [Next Step](#).
 - For each system, select from the following 2 options:
 - Enable Capability Deployment.
 - Enable the Joule Icon in the Integrated System.
 - Click [Review](#).
 - Review the changes and click [Include](#).

To exclude a system from an existing formation:

- Log in to the [SAP BTP account](#) as global account administrator.
- In the left navigation pane of the global account, click [System Landscape](#).
- Under System Landscape, click [Formations](#).
- Find your Joule Formation.
- Click the relevant system in the formation.
- On the right panel, click [Exclude System](#).
- Click [Exclude](#) to confirm the selection.

Note

You can't change the consent for the Conversation Insights and the Integration Landscape once it is set.

Delete the Joule Formation

Note

The Joule subscription is not deleted upon the deletion of the formation. To delete the Joule subscription, refer to [Unsubscribe from Joule \[page 17\]](#). You must delete the Joule formation before deleting the Joule subscription to avoid any errors in the existing integration set up.

If you want to remove the Joule integration from the SAP solution(s), go to [System Landscape](#) in the SAP BTP Cockpit and find the formation that you had created. Click [Delete Formation](#). Once the formation is successfully deleted, the configurations that were created for each product as part of the provisioning process are deleted and the Joule icon is removed from the product UI.

Note

The Joule subscription is not deleted upon the deletion of the formation.

Constraints

The following constraint is applied when running the Joule booster:

- The following product combination is not currently supported:
 - SAP S/4HANA Cloud Public Edition and SAP S/4HANA Cloud Private Edition

Related Information

[Integration with SAP SuccessFactors \[page 24\]](#)

[Integration with SAP S/4HANA Cloud Private Edition \[page 32\]](#)

[Integration with SAP Integrated Product Development \[page 66\]](#)

3.5 Configure Trusted Domains in Identity Authentication

You need to add the domain of the integration product as trusted domain in the Identity Authentication.

Prerequisites

You are assigned the Manage Tenant Configuration role. For more information about how to assign administrator roles, see [Edit Administrator Authorizations](#).

Procedure

1. Sign in to the administration console for SAP Cloud Identity Services.
2. Under [Applications and Resources](#), choose the [Tenant Settings](#) tile.
At the top of the page, you can view the administrative and license relevant information of the tenant.
3. Under [Customization](#), choose the [Trusted Domains](#) list item
4. Press the [+ Add](#) button.
5. Type the URL of the trusted host in the field.
You can type either the full name of the host or you can use the wild card [*](#) function.

⚠ Caution

Be careful when using the wild card [*](#) function. Make sure that you trust the domain when you use it.

See the examples:

mycompany.ondemand.com

*.example.com

6. Save your changes.
If the operation is successful the system displays the message **Trusted Domains updated**.

For more information, see [Configure Trusted Domains](#).

3.6 Configure Trusted Domain in SAP BTP

You can add, configure and save newly created trusted domain.

1. Go to your subaccount and choose [► Security ► Settings ►](#).
2. Choose [+ \(Add\)](#).
3. Enter the host name for the trusted domain.
4. For example: `https://store.example.com`
5. All the trusted domains together, including their space separators, can't exceed 2048 characters.
6. Save your entries.

3.7 Constraints for Joule

All constraints applicable to Joule are listed in this section. These constraints will provide you better understandability of Joule.

The following constraints apply for Joule:

- Joule currently supports conversations in English only.
- A Single Joule subscription can be integrated into multiple SAP products like SAP SuccessFactors, SAP S/4HANA Cloud Private Edition, however, one subscription of the Joule application can be integrated with

only one tenant of an SAP product. This means you cannot include multiple systems of same product in one Joule formation. Data isolation happens at tenant level.

- If you have created the NavigationService destination manually, then the subscription of Joule must be created in the same subaccount where the subscription and service instance of SAP Build Work Zone exists. However, this constraint does not apply if the destination is set up automatically via the booster. For more information, see [Prerequisites \[page 12\]](#).
- Joule requires a trust configuration of type OpenID Connect with your Identity Authentication tenant. SAML based trust is not supported.
- The domain of the Identity Authentication tenant used for setting up trust configuration in SAP BTP subaccount should be same as the IAS domain for the SAP product.
- It is not possible to create trust to the same Identity Authentication tenant based on SAML and OIDC in parallel. If you have to set up SAML trust to your application, you can create the Joule subscription in a different subaccount and set up OIDC trust with the same IAS tenant that is linked with the application.

Note

The domain of the Identity Authentication tenant used for setting up trust configuration in the SAP BTP subaccount should be the same as the IAS domain used for integration with the parent SAP product.

- Joule uses SAP Build Work Zone, standard edition with foundation plan for navigational scenarios. This implies that Identity Provisioning Service (IPS) must be enabled and configured for user assignment to the navigation content.
- Users can send up to three thousand requests per minute to Joule. Once the rate limit is reached, any subsequent requests will return an error code until enough time has passed for the call count to drop below the limit.

3.8 Unsubscribe from Joule

This section provides instructions on how to unsubscribe and delete a Joule subscription and service instance. Use it to efficiently manage your subscriptions and free up resources when no longer needed.

If you no longer need the Joule subscription or the service instance, you can delete them.

Upon unsubscribing from Joule, all new configurations will be deleted. All services that are part of Joule setup will be unsubscribed, and their persisted data will be deleted. Consequently, you will not be able to access your assistant and the chat data.

Note

Once you delete the subscription, all your tenant data is deleted. Consequently, you will not be able to access your assistant and the chat data.

Deleting the Joule Subscription

1. Open the [Cockpit](#) of your subaccount.

2. Open your [Subaccount](#) tile and click [Instances and Subscriptions](#) in the navigation pane on the left.
3. Find the application Joule under [Subscriptions](#).
4. Click [Delete](#) under the actions menu (•••) on the right.
The subscription is deleted.

Caution

After you have deleted the subscription of Joule, you may not be able to re-subscribe in the same subaccount. Create a subscription in another subaccount in which you haven't yet subscribed to Joule or raise a ticket to CA-DAS-T0.

4 Integrating Your SAP Solution

Use the following table to search for your SAP Solution, navigate the solution specific integration guide, and integrate your solution with Joule.

Integrating your solution

SAP Solution	Integration Guide
SAP Build Work Zone, standard edition - foundation plan	Integrating Content with SAP Build Work Zone standard edition - foundation plan [page 19]
SAP Build Work Zone, standard edition	Integration with SAP Build Work Zone, standard edition [page 22]
SAP Start	Integration with SAP Start [page 23]
SAP SuccessFactors	Integration with SAP SuccessFactors [page 24]
SAP S/4HANA Private Cloud	Integration with SAP S/4HANA Cloud Private Edition [page 32]
SAP S/4HANA Cloud Public Edition	Integration with SAP S/4HANA Cloud Public Edition [page 50]
SAP Integrated Product Development	Integration with SAP Integrated Product Development [page 66]
SAP Integrated Business Planning for Supply Chain	Integration with SAP Integrated Business Planning for Supply Chain [page 66]

4.1 Integrating Content with SAP Build Work Zone standard edition - foundation plan

An overview of the process for integrating content to the SAP Build Work Zone Foundation.

To support navigation to different apps from Joule and handle role based access scenarios, you need to configure the navigation service of SAP Build Work Zone. This allows resolution of intent based navigation targets that are defined in the backend. The content needs to be exposed by the SAP product (source system) and consumed by SAP Build Work Zone (target system). This process is composed of a content exposure flow and a content consumption flow.

Note

For integrating content from cloud solutions, all systems must operate under the same domain.

Content Exposure

The content exposure involves setting up the communication between the content provider system and the SAP Build Work Zone, and exposing the roles that you want to integrate. The steps required for content exposure vary for each SAP product. Please refer to the product specific integration section.

Content Consumption

After completing the content exposure, you need to set up the content consumption, at the end of which it will be possible to transfer content from SAP product to SAP Build Work Zone. You also need to configure a design-time destination from which the SAP Build Work Zone will fetch the exposed content. The following steps need to be performed for all integrations that support navigational scenarios.

- [Create SAP Build Work Zone Application and Instance \[page 20\]](#)

Related Information

[Integration with SAP SuccessFactors \[page 24\]](#)

[Integration with SAP S/4HANA Cloud Private Edition \[page 32\]](#)

4.1.1 Create SAP Build Work Zone Application and Instance

Steps to create a subscription and instance of SAP Build Work Zone, standard edition.

Make sure that you have the entitlement for SAP Build Work Zone, standard edition (foundation plan) for both application and service added in your global account. For more information, see [Prerequisites \[page 7\]](#).

Note

If your subaccount already has the set up for SAP Build Work Zone system, standard edition (standard plan), you can skip these entitlements and continue to use the same and add the missing configurations.

Create a Subscription for SAP Build Work Zone, standard edition

1. In the [SAP BTP Cockpit](#), open your subaccount.
2. Go to ► [Services](#) ► [Service Marketplace](#) ► in the left navigation pane.
3. Search for SAP Build Work Zone, standard edition, and click the tile.
4. In the top right corner, click [Create](#).
5. In the window that opens, keep all the default settings and click [Create](#).

6. Assign users to the **Launchpad_Admin** role collection. This role collection enables you to perform all the administration tasks for a SAP Build Work Zone, standard edition site, and is available out-of-the-box.
 1. Go to ► [Security](#) ► [Role Collections](#) ► in the left navigation pane.
 2. Search for the role collection **Launchpad_Admin**.
 3. Open the role collection for editing. Click the arrow at the far right, and in the role collection screen, click [Edit](#).
 4. In the [Users](#) section of the screen, search for your email and add it to the list of users by clicking the + button. Save your changes.

Create an Instance for SAP Build Work Zone, standard edition

1. Go to ► [Services](#) ► [Service Marketplace](#) ► in the left navigation pane.
2. Search for SAP Build Work Zone, standard edition, and click the tile.
3. In the top right corner, click [Create](#) and enter the details of the new service instance.

Note

You must enable Cloud Foundry and create a space before you create a service instance. For more information, see [Creating Service Instances in Cloud Foundry](#).

4. In the window that opens, keep all the default settings and click [Create](#).
5. To create a service key, still in the [Instances and Subscriptions](#) screen, click the ⋮ ([Actions](#)) next to the service instance entry in the table, and create a service key. For more information, see [Creating Service Keys in Cloud Foundry](#).

Related Information

[SAP Build Work Zone, standard edition](#)

4.1.2 Configure Destination to Use Navigation Service

Steps to configure the destination for using the navigation service.

You need to configure a destination with the name **NavigationService** in the subaccount with Joule set up to use the navigation service and ensure that these targets are resolved at runtime.

1. Go to your subaccount for Joule set up and choose ► [Connectivity](#) ► [Destinations](#) ► and create new destination with the following information:

Field	Value
Name	NavigationService

Field	Value
Type	HTTP
URL	portal-service URL from the service key created for the service instance of SAP Build Work Zone, standard edition.
Proxy Type	Internet
Authentication	OAuth2UserTokenExchange
Client ID	Client ID from the service key created for the service instance of SAP Build Work Zone, standard edition.
Client Secret	Client Secret from the service key created for the service instance of SAP Build Work Zone, standard edition.
Token Service URL Type	Dedicated
Token Service URL	https://<uaa url>/oauth/token

2. In the [Additional Properties](#) section, enter the following information:

Field	Value
Use default JDK trust store	Enable this option.

3. Save your changes.

4.2 Integration with SAP Build Work Zone, standard edition

Prerequisites

- You've switched to SAP Cloud Identity Services - Identity Authentication as Joule leverages Identity Authentication for user login. If you're not yet using Identity Authentication, please switch over. For more information, see [Switching to SAP Cloud Identity Services - Identity Authentication](#).
- Before completing the Joule onboarding tasks, you'll need to ensure that you've met the required prerequisites for both SAP Build Work Zone, standard edition and for Joule. For a list of these prerequisites, please refer to: [Prerequisites \[page 7\]](#).

Note

The trust is already created, so you can skip this particular step in the list of prerequisites.

- The Joule booster has been run. For more information, see [Run the Joule Booster \[page 12\]](#).

Setting Up the Joule Integration

1. Once the Joule booster has been run, and the integrations between SAP Build Work Zone, standard edition and Joule is complete, a setting becomes available in the [Site Settings](#) screen of the site.
2. As an administrator, you can then go to the [Site Settings](#) screen and enable the Joule setting.
3. Once enabled, the Joule icon is displayed in the header of your runtime site.

ⓘ Note

If you plan to set up Joule in another subaccount other than the SAP Build Work Zone, standard edition subaccount, then you need to ensure that for both subaccounts trust is established to the same SAP Cloud Identity Services - Identity Authentication tenant.

Restrictions

To view a list of restrictions and constraints when using Joule, see [Constraints for Joule \[page 16\]](#).

4.3 Integration with SAP Start

Prerequisites

- You have already run the manual setup for SAP Start, to obtain a subscription to SAP Build Work Zone, standard edition.
For more information, see [Manual Setup](#).

ⓘ Note

If you run the automated set up for SAP Start, you can't select an existing subaccount. It is recommended to run the manual set up for SAP Start and select an existing subaccount.

- SAP Build Work Zone, standard edition is configured to use SAP Cloud Identity Services - Identity Authentication.
For more information, see [Switching to SAP Cloud Identity Services - Identity Authentication](#).

Note

If you plan to set up Joule in another subaccount other than the SAP Build Work Zone, standard edition subaccount, then you need to ensure that for both subaccounts trust is established to the same SAP Cloud Identity Services - Identity Authentication tenant.

Procedure

1. Run the Joule booster.
For more information, see [Run the Booster](#).

Note

In steps 3 and 4 of the booster, you don't need to select anything from the [Products](#) and [Capability Packages](#) drop-down lists, respectively.

2. Add the Joule system created by the booster to the SAP Start formation:
 1. In your global account, select [System Landscape](#) from the left panel, and select the [Formations](#) tab.
 2. Search for your formation of type [Integration with SAP Start](#) and click [Include Systems](#) (in the top right corner of the formation).
 3. Select the Joule system to include in the formation. This system has your subaccount in the [Discovery](#) column and [Joule](#) in the [System Type](#) column.
 4. Click [Next Step](#).
 5. Click [Include](#).

As a result, the Joule system is added to the formation's list of systems.

You can now use Joule in SAP Start by clicking the Joule icon that was added to the header.

Restrictions

To view a list of restrictions and constraints when using Joule, see [Constraints for Joule](#).

4.4 Integration with SAP SuccessFactors

Steps required to integrate Joule in SAP SuccessFactors.

Prerequisites

- You have fulfilled the necessary prerequisites as mentioned in [Prerequisites \[page 7\]](#).

- You have integrated SAP SuccessFactors with Identity Authentication as Joule leverages the IAS setup of the integrated SAP product for user login. For more information, see [Getting Started with Identity Authentication and SAP SuccessFactors](#).
- You have created a service instance and generated a service key for SAP Build Work Zone, standard edition (foundation plan). For more information, see [Create SAP Build Work Zone Application and Instance \[page 20\]](#).

Integration Steps

The integration process involves the following steps:

- [Register an SAP SuccessFactors System in a Global Account in SAP BTP](#)
- [Run the Joule Booster \[page 12\]](#) with SAP SuccessFactors as an option in the *Select Integrations* screen. You can select other products in combination with SAP SuccessFactors if you have appropriate licenses and entitlements.
- [Create SAP Build Work Zone Application and Instance \[page 20\]](#)
- [Consume Content from SAP SuccessFactors \[page 25\]](#)
- [Configure Identity Provisioning Service \[page 28\]](#)

4.4.1 Consume Content from SAP SuccessFactors

SAP SuccessFactors provides a REST API to push CDM content to SAP Start. You need to configure a design time destination and a runtime destination in the SAP BTP subaccount, and a content channel in the SAP Build Work Zone Content Manager to push the content to the SAP Start site.

Create a Design Time Destination

Create a design-time destination on SAP BTP to access the CDM content API from SAP SuccessFactors.

Note

Accessing SAP SuccessFactors APIs using Basic Authentication has been deprecated. You can create certificate based destinations. For more information, see [Deprecation of HTTP Basic Authentication for APIs](#).

1. Log into your SAP BTP cockpit and go to your subaccount for Joule set up.
2. Choose  [Connectivity](#)  and create new destination with the following information:

Field	Value
Name	LPS_SFSF_dt

Field	Value
Type	HTTP
URL	https://<tenant API URL>/rest/servicesfoundation/ sfcdmcontentservice/v1/SFCDMContent
Proxy Type	Internet
Authentication	Basic Authentication
User	Enter your SAP SuccessFactors username with oData API access and company in the format of username@COM- PANY
Password	Enter the password for your SAP SuccessFactors

3. In the *Additional Properties* section, enter the following information:

Field	Value
Use default JDK trust store	Enable this option
HTML5.Dynamic Destination	true

4. Save your changes.

Update the Runtime Destination

LPS_SFSSF_rt destination is automatically created when you run the Joule booster. It has the following configurations:

Field	Value
Name	LPS_SFSSF_rt
Proxy Type	Internet
Authentication	NoAuthentication
Type	HTTP
URL	Your SAP SuccessFactors tenant URL
Additional Properties	
Field	Value
sfsf.companyID	SAP SuccessFactors company

However, you may need to update the destination in the following scenarios:

- If you are using SAP Build Work Zone foundation plan (not standard plan), add the following property in the [Additional Properties](#) section by typing `sap-start` in the field. It is not available in the drop-down menu.

Field	Value
sap-start	true

- If your SAP SuccessFactors tenant is already migrated to `cloud.sap` (SAP super domain), update the URL field in the destination to use new super domain, for example, `https://sfsf.cloud.sap`.

Add a Content Provider to Consume CDM Content

Add a new content provider to your SAP Start site to consume the CDM content from SAP SuccessFactors. For information on accessing the [Site Manager](#), see section *Open and Manage SAP Build Work Zone Application* in this [tutorial](#).

1. Go to **Services > Service Marketplace** in the left navigation pane.
2. Click the SAP Build Work Zone, standard edition application subscription hyperlink to open the application.
3. Go to the [Site Manager](#) page and open the [Channel Manager](#) tab.
4. Choose **+ New > Content Provider** to add a new content provider.
5. Enter the following information:

Field	Value
Title	Enter a name for the content provider (recommended sfsf)
Description	Enter a description for the content provider
ID	Any unique ID (recommended sfsf)
Design-Time Destination	Select the design time destination LPS_SF_SF_dt
Runtime Destination	Select the runtime destination LPS_SF_SF_rt
Runtime Destination for Dynamic Data	Select Use default runtime destination
Automatically add all content items to subaccount	True
Use the Identity Provisioning service to provision user authorizations	True

6. Save the entry.

4.4.2 Configure Identity Provisioning Service

Steps to configure Identity Provisioning service.

Navigation service of SAP Build Work Zone, standard edition service uses Identity Provisioning service (IPS) that is used to provision identities and their authorizations between source and target systems.

This section describes the steps to configure the source system (SAP SuccessFactors) and target systems (Identity Authentication and SAP Build Work Zone, standard edition) in the Identity Provisioning of your IAS application user interface. For some customers, SAP SuccessFactors and the Identity Authentication systems are already configured as the source and target system by the [Upgrade Center](#).

The following configurations are required to provision:

1. User details like user email, Global User ID and group memberships from SAP SuccessFactors to SAP Build Work Zone, standard edition.
2. User roles as groups from SAP SuccessFactors (source system) to SAP Build Work Zone, standard edition with role ID as external ID and group memberships.

Note

You must have the administrative access to your Identity Provisioning tenant and you need to enable the [Manage Identity Provisioning](#) role for your user. For more information, see [Manage Authorizations in SAP Cloud Identity Infrastructure](#).

3. Login to your Identity Authentication tenant.
4. Navigate to [Identity Provisioning](#) [Source System](#).
5. Select the source system configured for SAP SuccessFactors from the left panel and go to the [Transformations](#) tab. You can switch to the JSON editor by choosing the code-bracket icon.

Note

Existing Identity Authentication customers must switch from the ODATA API to the SCIM API for use with Identity Provisioning. For more information, see [Upgrade from ODATA IPS Connector to SCIM IPS Connector with SAP SuccessFactors HXM Suite](#).

6. Ensure that the [group](#) entity has following configuration:

Property	Value	Description
Ignore	false	Ensures groups SCIM entity is considered during the provisioning jobs
Mappings	Sample Code <pre>{ "sourcePath": "\$.id", "targetPath": "\$.externalId",}</pre>	Ensures the source ID field of the SCIM entity groups is set to externalId

7. Under the [Properties](#) tab, ensure that the field `sf.user.filter` is configured to fetch all the required and valid users.

8. Follow these steps if you don't want the groups to be provisioned in IAS, else skip this step and go to step 7.
 1. Navigate to ► [Identity Provisioning](#) ► [Target System](#) ►.
 2. Select the target system configured for Identity Authentication from the left panel and go to the [Transformations](#) tab. You can switch to the JSON editor by choosing the code-bracket icon.
 3. Ensure that the [group](#) entity has following configuration:

Property	Value	Description
Ignore	true	Ensures groups SCIM entity is ignored during the provisioning jobs

9. Create a new target system with following information:

Field	Value
Type	SAP Build Work Zone, standard edition
Name	Any meaningful name (Work Zone-Target)
Description	Any meaningful description
Source System	Select SuccessFactors source system

10. Select the target system that you created from the left panel and go to the [Transformations](#) tab. You can switch to the JSON editor by choosing the code-bracket icon.
11. Choose [Edit](#). Add the following attribute mapping for the [group](#) entity.

Property	Value	Description
Mapping	 Sample Code <pre>{ "sourcePath": "\$.externalId", "targetPath": "\$.externalId", },</pre>	Ensures the externalId field of the SCIM entity groups is set to externalId

12. Switch to [Properties](#) tab and add or confirm if the following properties are available using service key generated earlier for the SAP Build Work Zone, standard edition (foundation or standard plan):

Field	Value
URL	portal-service field value under endpoints node from the service key
Authentication	BasicAuthentication

Field	Value
User	<code>clientId</code> field value under <code>uaa</code> node from the service key
Password	<code>clientsecret</code> field value under <code>uaa</code> node from the service key
ProxyType	Internet
Type	HTTP
OAuth2TokenServiceURL	<code>https://<uaa url>/oauth/token</code>
<code>ips.trace.failed.entity.content</code>	false
<code>cflp.user.unique.attribute</code>	<code>emails[0].value,</code> <code>['urn:ietf:params:scim:schemas:extension:2.0:mapping']</code> <code>['providerId'],externalId</code>
<code>cflp.support.bulk.operation</code>	false
<code>cflp.providerId</code>	ID field value for content channel configured for SAP SuccessFactors in SAP Build Work Zone
<code>cflp.group.unique.attribute</code>	<code>externalId,</code> <code>['urn:ietf:params:scim:schemas:extension:2.0:mapping']['providerId']</code>
<code>cflp.bulk.operations.max.count</code>	100

Note

Add the properties `cflp.support.bulk.operation` (value as `true`) and `cflp.bulk.operations.max.count` if your source system has too many entries, else you can omit them. For more information, see [List of Properties](#).

13. Navigate to ► [Identity Provisioning](#) ► [Source System](#) ►.
14. Select the source system configured for SAP SuccessFactors and switch to [Jobs](#) tab.
15. Run a [Read](#) or a [Resync](#) job as per your requirements to provision SAP SuccessFactors users and roles to SAP Build Work Zone ([Navigation Service](#)).

Note

1. We recommended to run Identity provisioning service (IPS) provisioning job on scheduled basis with interval less than or equal to twenty four hours for permissions to take effect on time and overlap with [Refresh Synthetic Group Data](#) job in SAP SuccessFactors.
2. Depending on the job schedules and configuration, it may take twenty four to forty eight hours for the roles to reflect in the navigation service from SAP SuccessFactors.

3. You may need to run [Refresh Synthetic Group Data](#) and IPS job in SAP SuccessFactors and IPS job to sync the roles on adhoc basis or as needed.

4.4.3 Enabling Joule in SAP SuccessFactors

Administrators can enable Joule and grant the role-based permission to SAP SuccessFactors users who need to access the copilot.

Procedure

1. Follow the initial set up procedure linked in [Related Information](#) to complete initial settings on SAP Business Technology Platform.

After you complete the settings, Joule is enabled in your instance. Then, you need to grant the role-based permission to SAP SuccessFactors users.

2. At [SAP SuccessFactors](#) > [Admin Center](#) > [Manage Permission Groups](#), create a permission group for the users who need to access Joule.

All users in this permission group can access Joule after you grant the required role-based permission to the group.

3. At [Admin Center](#) > [Manage Permission Roles](#), create new permission roles for Joule users.

→ Tip

You can also update existing permission roles. For more details about how to create a permission role, refer to [Permission Roles](#).

- a. Choose [Create New](#).
- b. On the [Permission Role Detail](#) page, choose [Permission](#).
- c. In the [Permission settings](#) dialogue box, grant the [User Permissions](#) > [General User Permission](#) > [Access to Joule](#) permission and other permissions if needed to the permission role, and then choose [Done](#) to save your changes.

📌 Note

To access use cases, make sure that users have the [User Permissions](#) > [General User Permission](#) > [User Search](#) permission.

- d. On the [Permission Role Detail](#) page, choose [Grant this role to](#) > [Add](#) and grant this role to the permission group that you've configured for Joule users in Step 2.
 - e. Choose [Done](#).
4. Save all your changes.

If the changes don't take effect within 10 minutes, you can manually refresh the permissions according to [Refreshing RBP after Changes in Provisioning Configuration](#).

Note

For use cases with the navigation capability, users may not be able to see the navigation links immediately after they are granted the required permissions. Please expect 1-2 days for such permission changes to take effect.

5. As a prerequisite to use each case for Employee Central Quick Actions, you must enable this feature and configure the corresponding templates.

→ Tip

For more information about the configuration, refer to [Employee Central Quick Actions](#).

Results

Joule is now ready to use.

Related Information

[Initial Setup](#)

[What Are Role-Based Permissions?](#)

4.5 Integration with SAP S/4HANA Cloud Private Edition

This section describes the steps to enable the Joule Communication Scenario on your SAP S/4HANA Private Cloud instance and integrate Joule Web Client plugin in your SAP Fiori Launchpad, allowing your users to interact with Joule.

Prerequisites

- You have carried out the steps described in [Initial Setup \[page 6\]](#).
- You have referred to the [Security](#) related information for Joule.
- You have fulfilled the necessary prerequisites as mentioned in [Prerequisites \[page 7\]](#).
- You have integrated the SAP S/4HANA Cloud Private Edition with Identity Authentication as Joule leverages the IAS setup of the SAP product for user login.
- You have created a service instance and generated a service key for SAP Build Work Zone, standard edition (foundation plan). For more information, see [Create SAP Build Work Zone Application and Instance \[page 20\]](#).

- You have added the entitlement for the [connectivity](#) plan of the Cloud Identity services in your subaccount.

Integration Steps

1. [Configure Trust to the Identity Authentication Tenant \[page 10\]](#)
2. [Configure Trusted Domain in SAP BTP \[page 16\]](#)
3. [Configure User Attributes for Joule from the Identity Directory \[page 11\]](#)
4. [Configure Trusted Domains in Identity Authentication \[page 15\]](#)
5. [Expose SAP Fiori Launchpad Content to SAP BTP \[page 35\]](#)
6. [Set Up SAP S/4HANA Cloud Private Edition as a Content Provider \[page 33\]](#)
7. [Integrating Content with SAP Build Work Zone standard edition - foundation plan \[page 19\]](#)
8. [Run the Joule Booster \[page 12\]](#)
9. [Configure Destinations \[page 36\]](#)
10. [Configure Identity Provisioning Service \[page 41\]](#)
11. [Configure and Activate Joule Plug-In in a Target Mapping \[page 48\]](#)

4.5.1 Set Up SAP S/4HANA Cloud Private Edition as a Content Provider

Set up SAP S/4HANA Cloud Private Edition as a content provider for SAP Build Work Zone, standard edition using the SAP Cloud Connector.

⚠ Caution

SAP Build Work Zone, standard edition automatically uses SAP Cloud Connector as a proxy to any system connected to it. All paths that you configure in your cloud connector will be reachable via the public internet. Please make sure that you only configure the necessary paths and that your SAP S/4HANA Cloud Private Edition system is properly secured to prevent unauthorized access.

1. Establish trust between your subaccount and your SAP S/4HANA Cloud Private Edition system.
For more information, see [Cloud Connector](#).
Take note of the following:
 - The section about [Configure Principal Propagation](#).
 - Disable the [System Certificate for Logon](#) setting to enable principal propagation and basic authentication via the SAP Cloud Connector.
For more information, see [Expose Intranet Systems](#).
2. In your SAP Cloud Connector Administration UI, open the [Cloud To On-Premise](#) tab.
3. Create a new system mapping and expose the following paths for the HTTPS protocol:

Path to fetch...	Value
Card manifests and i18n files	/sap/bc/lrep/

Path to fetch...	Value
The UI5 App Index	/sap/bc/ui2/app_index/
CDM3 content	/sap/bc/ui2/cdm3/
Data from OData services	/sap/opu/odata/
Data from OData V4 services	/sap/opu/odata4/

Note

The paths in the table above are for a default configuration, that covers most use cases. However, from a security perspective, we recommend to restrict the paths further, according to the services that are used in your use case.

For more information, see [Set Up Cloud Connector](#).

4. Create a new mapping for the integration with the Identity Provisioning Service using the following values:

Field	Value
<i>Back-end Type</i>	ABAP System
<i>Protocol</i>	RFC or RFC SNC
<i>Connection Type</i>	With load balancing (system ID and message server)
<i>Message Server</i>	<Your SAP S/4HANA Cloud Private Edition message server host, e.g. ldcsuct>
<i>System ID</i>	<Your SAP S/4HANA Cloud Private Edition system ID, e.g. UCT>
<i>SAProuter</i>	Leave empty.
<i>Virtual Message Server</i>	ldcsuct
<i>Virtual System ID</i>	UCT

Note

Using the values in the table above, the value of the internal host would be: **ldcsuct:sapmsUCT**

5. With the *Naming Policy* set to *Exact Name* in the Identity Provisioning service, allow access to the following resources:

- BAPI_USER_GETLIST
- BAPI_USER_GET_DETAIL
- PRGN_ACTIVITY_GROUPS_LOAD_RFC
- PRGN_ROLE_GETLIST

For more information, see [SAP Application Server ABAP](#).

6. Open your SAP S/4HANA Cloud Private Edition system to preview and expose your content.

Note

- *Exposure Version: Version 2*

- To check that your content exposure was successful, you can use the SAP Gateway Client (transaction /IWFND/GW_CLIENT) as follows: Send a GET request to /sap/bc/http/ui2/flp_content_exposure and check that you receive as a response the status code 200.

For more information, see [Manage Launchpad Content for Exposure](#).

7. To create a technical service user, open transaction SU01 and as the *User Type* select *Service*.
8. Make sure that the technical service user fulfills the prerequisites in [Manage Launchpad Content for Exposure](#).
9. In the *My Home Settings*, make sure that the business users are assigned the value: *Insights Cards*.

Related Information

[Integrate SAP S/4HANA Cloud Private Edition](#)

4.5.2 Expose SAP Fiori Launchpad Content to SAP BTP

Administrators can expose SAP Fiori launchpad content to SAP BTP by choosing specific roles. The related content, such as groups, catalogs, pages, or spaces can then be consumed on SAP BTP. For information, see [Manage Launchpad Content for Exposure](#).

Procedure

Expose your SAP Fiori launchpad content:

1. Launch transaction /UI2/CDM3_EXP_SCOPE.
2. Choose the exposure version you would like to use. See [Content Exposure Versions Overview](#).

Note

Each exposure version uses a dedicated repository, allowing you to run, schedule, and use both version 1 and version 2 simultaneously.

3. Choose which roles you want to provide to SAP BTP. You have the following options:
 - *All roles with launchpad content* - By default, all roles are exposed.
 - *Selected roles in areas (all selected roles in all areas will be exposed)* - Select specific roles or bundle several roles in a specific area. See [Select Roles for Content Exposure](#).

Note

All roles contained in the role selections of all areas are used for the exposure. To add additional roles to the exposed content, simply include these roles in the existing role selection and execute the exposure. Conversely, to remove specific roles from the exposed content, simply exclude these roles from the existing role selection and execute the exposure.

Note

Derived roles are not supported directly. If you want to expose a derived role, choose the leading role containing the launchpad content and add it to the role selection. The derived role will be exposed indirectly as a related role assignment.

4. Optional: Click [Preview Content](#) or [Preview Area Content](#) to see a preview of the selected content. This might help you to see if you have selected all your required content and if the content contains any errors or warnings that you might need to fix before exposing.
5. Save your defined roles by clicking on [Save Selected Roles](#). The role selection must be saved first before exposing. Otherwise you will expose the **last** saved role selection.
6. Check whether your content contains any errors by either clicking [View Log](#) or [Preview Content](#)  [Display Log](#) . Note that only content without errors can be exposed.
7. Expose your launchpad content:
 - Schedule a job to expose your content on a regular basis. The exposure process will use the saved configuration. See [Schedule Content Exposure](#).
 - Click [Expose Content](#) to manually trigger the exposure of the saved configuration.

Related Information

[Manage Launchpad Content for Exposure](#)

4.5.3 Configure Destinations

This document provides you with the information to configure a design-time destination, runtime destination, and single sign-on (SSO).

To integrate the content you exposed on the SAP S/4HANA Private Cloud system into SAP Build Work Zone, standard edition, you need to configure a design-time destination to define the location from which to fetch the exposed content. In addition, you configure the runtime destinations to obtain the resources needed to run the federated apps in runtime. For more information, see [Configure Destinations](#).

Design-Time Destination

In the SAP BTP cockpit, in your subaccount, navigate to [Connectivity](#)  [Destinations](#) .

Use the following properties to define the design-time destination:

Property	Value
Name	We recommend to add the suffix dt to the name. Note The name must contain only lowercase letters.
Type	<i>HTTP</i>
Description	Enter a description
URL	The URL of the Virtual Host of your Cloud Connector: https://<host>:<port>/sap/bc/ui2/cdm3/entities
Proxy Type	<i>OnPremise</i>
Authentication Method	Note The selected method must also be supported on the side of the content provider. The following methods is supported:<i>Basic Authentication</i>. To connect to the backend, use a technical user with permission to read the CDM. Note <i>Principal Propagation</i> is not supported.
sap-client (additional property)	The client number of the ABAP system. For example: 120

Runtime Destination

The runtime destination defines the location from which to obtain the resources needed to run the federated apps in runtime.

Property	Value
Name	We recommend to add the suffix rt to the name. Note The name must contain only lowercase letters and must not contain the underscore (_) character.
Type	<i>HTTP</i>

Property	Value
Description	Enter a description
URL	The URL of the Virtual Host of the system that is set up in your Cloud Connector. You find this value in the Cloud Connector Administration tool - select Cloud to On-Premise and in Access Control, use the value in the Virtual Host column. For example: <code>http://qky:44300</code> For more information, see Set Up Cloud Connector. Note For Web Dynpro ABAP applications, if they include SAP Business Client (<code>CompatibilityMode=TRUE</code>), it is necessary to configure the value of the Cloud Connector virtual host to be the same as the full path. For more information, see the settings for WDA Apps Integration Mode in Maintaining Launchpad App Descriptor Items.
Proxy Type	OnPremise
Authentication Method	Depends on the authentication method in use. To use single sign on (SSO), select Principal Propagation .

Add the following additional properties:

Property	Value
HTML5.DynamicDestination	Add this property and set its value to true when you are creating a destination for a dynamic tile. Note To enable CSRF protection via the approuter, set the <code>CEP.EnableCsrfProtection</code> destination property to true for all dynamic destinations where their respective backend doesn't handle CSRF protection. If your backend is already protected, there's no need to add this property.
sap-platform	ABAP
sap-client	The client number of the ABAP system. For example: 120
sap-sysid	System ID of the SAP system (also referred to as SID). For example: QKY
launchpad.wa.productId, launchpad.wa.productVersion	To enable SAP Companion content (on-screen help) for the SAP S/4HANA apps running on this system (destination), you need to configure the product and version of the SAP Companion content that corresponds to the SAP S/4HANA apps. For more information, see Activating SAP Companion Content .

Property	Value
sap-provider-label	Add this property to provide a user-friendly display name for the system of a destination. This system label is used in various runtime features, such as: - It can be displayed directly on each tile of an SAP S/4HANA app, if this setting was enabled in the Site Settings. - It is shown in the drop-down list of the search results (only when using the Spaces and Page - New Experience view mode) and in the Search Results page. - In the Source System field in the user Default Values option, located under the User Actions menu - Settings Default Values. - In the App Finder.
sap-service	A concatenated string that contains 4 characters: the first 2 characters are "32"; the last 2 characters are the instance number of the ABAP application server or the SAP system number. For example: 3200

Runtime Destination for Dynamic Data

A dynamic tile displays data that is updated during runtime. To retrieve the dynamic data, it is necessary to define an additional runtime destination for dynamic data. When defining a content provider in the Channel Manager, the default value for the dynamic data destination is the same one as the default runtime destination. However, you can select a different destination than the default one, if required.

Destination for Retrieving Nominations via the SAP Cloud Connector

Configure an additional destination (for example privatecloud) for retrieving nominations via the SAP Cloud Connector, using the following values:

Field	Value
<i>Name</i>	privatecloud
<i>Type</i>	HTTP
<i>URL</i>	<Virtual host and port from SAP Cloud Connector>
<i>ProxyType</i>	OnPremise
<i>Authentication</i>	PrincipalPropagation
<i>Location ID</i>	<ID of the SAP Cloud Connector, if configured>

Add the following additional properties with their values (case sensitive):

Property	Value
<i>HTML5.DynamicDestination</i>	true
<i>nameIdFormat</i>	tc:SAML:1.1:nameid-format:emailAddress

Property	Value
<i>sap-card-nominations-path</i>	/sap/opu/odata4/ui2/insights_srv/srvd/ui2/insights_cards_read_srv/0001/CEP_Cards?\$expand=DescriptorResources

Create a Content Provider

In SAP Build Work Zone, standard edition, in the [Channel Manager](#), define the SAP S/4HANA Private Cloud system as content provider. For more information, see [Manage Content Providers](#).

1. In the SAP BTP cockpit, log on to your subaccount as administrator.
2. In your subaccount, go to [Instances and Subscriptions](#) and launch SAP Build Work Zone, standard edition.
3. In SAP Build Work Zone, standard edition, go to [Channel Manager](#).
4. Choose **+ New** to add a new content provider.

Field	Value
<i>Title</i>	SAP S/4HANA Cloud Private Edition or any other meaningful name
<i>ID</i>	<A unique ID>
<i>Design-Time Destination</i>	Select the design time destination
<i>Runtime Destination</i>	Select the runtime destination
<i>Runtime Destination for Dynamic Data</i>	The runtime destination for retrieving dynamic data to display on dynamic tiles. By default, the default runtime destination is used.
<i>Automatically add all content items to subaccount</i>	true
<i>Use the Identity Provisioning service to provision user authorizations</i>	true
<i>Include group and catalog assignments to roles</i>	Use this toggle switch depending on how the provider is modeled • Disable this feature to include all groups and catalogs in this site, without considering their assignment to roles. • Enable this feature to include only groups and catalogs in this site, that have been directly assigned to roles. For more information, see Defining Optional Relations Between Roles and Groups/Catalogs.

A new row is added to the Content Channels table, with the status “Creating...”.

When the content provider is ready, the status changes to “Created”.

Click the Report link to see the number of roles, apps, groups, catalogs, spaces, pages, and URL templates that are included in the content provider.

The content provider is now visible also in the Content Explorer.

Single Sign-On Destination

- In the SAP BTP cockpit, log on to your subaccount for Joule as administrator.
- In the navigation area, choose ► [Connectivity](#) ► [Destinations](#) .

To create the destination, choose [New Destination](#) and enter the following information:

Property	Description/Value
Name	S4HANA_PCE_SSO
Type	HTTP
Description	Enter a description.
URL	The URL of the Virtual Host of your Cloud Connector : <code>https://<host>:<port></code>
Proxy Type	OnPremise
Authentication	Principal Propagation is supported.
sap-client (additional property)	The client number of the ABAP system.

4.5.4 Configure Identity Provisioning Service

Configure the Identity Provisioning service to provision identities and their authorizations to SAP Build Work Zone, standard edition.

Procedure

1. In your SAP S/4HANA Cloud Private Edition system, open transaction SU01 and add the role SAP_BC_JSF_COMMUNICATION_R0 to the technical service user used for reading the content.

Note

2. Go to your BTP Account and navigate to [Instances and Subscriptions](#) and subscribe to [Cloud Identity Services](#) with plan [connectivity](#).
3. Navigate to ► [Connectivity](#) ► [Destinations](#) , and create a new destination for the Identity Provisioning integration.

For more information, see [SAP Application Server ABAP](#).

The following tables contain as an example the values used for a connection with load balancing:

Field	Value
Name	Name of the destination
Type	RFC

Field	Value
Description	Enter a description
ProxyType	OnPremise
User	<Username of your SAP S/4HANA Cloud Private Edition technical service user>
Password	<Password of your SAP S/4HANA Cloud Private Edition technical service user>
Authorization Type	CONFIGURED_USER
Location ID	<ID of the SAP Cloud Connector, if configured, example DLM_MAIL>

Add the following additional properties with their values (case sensitive):

Property	Value
<code>jco.client.client</code>	<Client of your SAP S/4HANA Cloud Private Edition system, example 950>
<code>jco.client.mshost</code>	<Virtual host of your SAP Cloud Connector configuration for Identity Provisioning service integration, until the colon, example uctclnt950rfc>
<code>jco.client.r3name</code>	<System ID of your SAP S/4HANA Cloud Private Edition system, example UCT>

4. Open your Identity Authentication service administrator console at: `<IAS domain>/admin`
5. Navigate to the [Identity Provisioning](#) section and open the [Source Systems](#) section.
6. Click **+ Add** to add a new Source System:
 1. Select [SAP Application Server ABAP](#) as the type.
 2. Give the source system a meaningful name.
 3. In the [Transformations](#) tab, click [Edit](#), use the `{..}` button to switch to JSON mode, and paste the following standard transformation:

Source Code

```
{
  "user": {
    "mappings": [
      {
        "sourcePath": "$.USERNAME",
        "targetVariable": "entityIdSourceSystem"
      },
      {
        "sourcePath": "$.USERNAME",
        "targetPath": "$.userName",
        "correlationAttribute": true
      },
      {
        "sourcePath": "$.ALIAS.USERALIAS",
        "targetPath": "$.externalId",
        "optional": true,
        "correlationAttribute": true
      }
    ]
  }
}
```

```

    {
      "sourcePath": "$.SAPUSER_UUID.SAP_UID",
      "targetPath": "$
['urn:ietf:params:scim:schemas:extension:sap:2.0:User']['userUuid']",
      "optional": true
    },
    {
      "targetPath": "$.schemas[0]",
      "constant": "urn:ietf:params:scim:schemas:core:2.0:User"
    },
    {
      "sourcePath": "$.ADDRESS.E_MAIL",
      "targetPath": "$.emails[0].value",
      "optional": true,
      "correlationAttribute": true
    },
    {
      "targetPath": "$.emails[0].primary",
      "condition": "$.ADDRESS.E_MAIL EMPTY false",
      "constant": true
    },
    {
      "targetPath": "$.emails[0].type",
      "condition": "$.ADDRESS.E_MAIL EMPTY false",
      "constant": "work"
    },
    {
      "sourcePath": "$.ADDRESS.FIRSTNAME",
      "targetPath": "$.name.givenName",
      "optional": true
    },
    {
      "sourcePath": "$.ADDRESS.LASTNAME",
      "targetPath": "$.name.familyName"
    },
    {
      "sourcePath": "$.ADDRESS.MIDDLENAME",
      "targetPath": "$.name.middleName",
      "optional": true
    },
    {
      "sourcePath": "$.ADDRESS.NICKNAME",
      "targetPath": "$.nickName",
      "optional": true
    },
    {
      "sourcePath": "$.ADDRESS.TITLE_P",
      "targetPath": "$.name.honorificPrefix",
      "optional": true
    },
    {
      "sourcePath": "$.ADDRESS.COUNTRY",
      "targetPath": "$.addresses[0].country",
      "optional": true
    },
    {
      "targetPath": "$.addresses[0].primary",
      "condition": "$.ADDRESS.COUNTRY EMPTY false",
      "constant": true
    },
    {
      "targetPath": "$.addresses[0].type",
      "condition": "$.ADDRESS.COUNTRY EMPTY false",
      "constant": "work"
    },
    {
      "sourcePath": "$.ADDRESS.TEL1_NUMBR",
      "targetPath": "$.phoneNumbers[0].value",

```

```

        "optional": true
    },
    {
        "targetPath": "$.phoneNumbers[0].primary",
        "condition": "$.ADDRESS.TEL1_NUMBR EMPTY false",
        "constant": true
    },
    {
        "targetPath": "$.phoneNumbers[0].type",
        "condition": "$.ADDRESS.TEL1_NUMBR EMPTY false",
        "constant": "work"
    },
    {
        "sourcePaths": [$.DEFAULTS.LANGU],
        "targetPath": $.locale,
        "valueMappings": [
            {
                "key": ["W"],
                "mappedValue": "bg"
            }
        ],
        "optional": true,
        "type": "valueMapping",
        "defaultValue": "en"
    },
    {
        "sourcePaths": [$.ADDRESS.LANGUP_ISO],
        "targetPath": $.preferredLanguage,
        "optional": true,
        "type": "valueMapping",
        "functions": [
            {
                "function": "toLowerCaseString"
            }
        ]
    },
    {
        "sourcePaths": [$.LOGONDATA.TZONE],
        "targetPath": $.timezone,
        "valueMappings": [
            {
                "key": ["EET"],
                "mappedValue": "Europe/Sofia"
            }
        ],
        "optional": true,
        "type": "valueMapping",
        "defaultValue": "Europe/Berlin"
    },
    {
        "targetPath": $.active,
        "constant": false
    },
    {
        "targetPath": $.active,
        "condition": "($.ISLOCKED.LOCAL_LOCK != 'L') &&
        ($.ISLOCKED.GLOB_LOCK != 'L') && ($.ISLOCKED.WRNG_LOGON != 'L')",
        "constant": true
    },
    {
        "sourcePath": $.ACTIVITYGROUPS[*].AGR_NAME,
        "targetPath": $.groups[?(@.value)],
        "optional": true,
        "preserveArrayWithSingleElement": true
    }
],
},
"group": {

```

```

"mappings": [
  {
    "sourcePath": "$.ROLE_NAME",
    "targetVariable": "entityIdSourceSystem"
  },
  {
    "sourcePath": "$.ROLE_NAME",
    "targetPath": "$.displayName",
    "functions": [
      {
        "function": "concatString",
        "condition": "'%abap.role.prefix%' !== 'null'",
        "prefix": "%abap.role.prefix%"
      }
    ]
  },
  {
    "targetPath": "$.schemas[0]",
    "constant": "urn:ietf:params:scim:schemas:core:2.0:Group"
  },
  {
    "sourcePath": "$.USERLIST[*].USERNAME",
    "targetPath": "$.members[?(@.value)]",
    "optional": true,
    "preserveArrayWithSingleElement": true
  }
]
}

```

7. Navigate to the [Identity Provisioning](#) section and open the [Target Systems](#) section.
8. Click [+ Add](#) to add a new Target System:
 1. Select SAP Build Work Zone, standard edition as the type.
 2. Give the target system a meaningful name.
 3. Select the source system you created.
 4. In the [Transformations](#) tab, click [Edit](#), use the [{..}](#) button to switch to JSON mode, and paste the following standard transformation:

Source Code

```

{
  "user": {
    "condition": "($.emails EMPTY false) && ($
['urn:ietf:params:scim:schemas:extension:sap:2.0:User'].userUuid EMPTY
false)",
    "skipOperations": ["update"],
    "mappings": [
      {
        "targetPath": "$.id",
        "sourceVariable": "entityIdTargetSystem"
      },
      {
        "targetPath": "$.schemas[0]",
        "constant": "urn:ietf:params:scim:schemas:core:2.0:User"
      },
      {
        "targetPath": "$
['urn:ietf:params:scim:schemas:extension:2.0:mapping']['providerId']",
        "constant": "%cflp.providerId%"
      },
      {
        "sourcePath": "$.emails[0].value",

```

```

        "targetPath": "$.emails[0].value",
        "condition": "$.emails[?(@.primary == true)].value == []",
        "optional": true
    },
    {
        "sourcePath": "$.emails[?(@.primary == true)].value",
        "targetPath": "$.emails[0].value",
        "condition": "$.emails[?(@.primary == true)].value != []",
        "optional": true,
        "preserveArrayWithSingleElement": true,
        "functions": [
            {
                "function": "elementAt",
                "index": 0
            }
        ]
    },
    {
        "targetPath": "$.emails[0].primary",
        "condition": "$.emails[0].length() > 0",
        "constant": true
    },
    {
        "sourcePath": "$
[urn:ietf:params:scim:schemas:extension:sap:2.0:User]['userUuid']",
        "targetPath": "$.externalId",
        "optional": true
    },
    {
        "sourcePath": "$.groups[*].value",
        "targetPath": "$.groups[?(@.value)]",
        "optional": true,
        "preserveArrayWithSingleElement": true,
        "functions": [
            {
                "function": "resolveEntityIds",
                "entityType": "group"
            }
        ]
    }
]
},
"group": {
    "mappings": [
        {
            "targetPath": "$.id",
            "sourceVariable": "entityIdTargetSystem"
        },
        {
            "targetPath": "$
[urn:ietf:params:scim:schemas:extension:2.0:mapping]['providerId']",
            "constant": "%cflp.providerId%"
        },
        {
            "targetPath": "$.schemas[0]",
            "constant": "urn:ietf:params:scim:schemas:core:2.0:Group"
        },
        {
            "targetPath": "$.schemas[1]",
            "constant": "urn:ietf:params:scim:schemas:core:2.0:mapping",
            "optional": true
        },
        {
            "sourcePath": "$.displayName",
            "targetPath": "$.externalId"
        }
    ],

```

```

{
  "sourcePath": "$.externalId",
  "targetPath": "$.externalId",
  "optional": true,
  "functions": [
    {
      "function": "replaceAllString",
      "regex": "(?i)(^pcd:)",
      "replacement": ""
    },
    {
      "function": "replaceString",
      "target": "/",
      "replacement": ":"
    },
    {
      "function": "replaceString",
      "target": "(",
      "replacement": "@"
    },
    {
      "function": "replaceString",
      "target": ")",
      "replacement": "+"
    }
  ]
},
{
  "sourcePath": "$.members[*].value",
  "targetPath": "$.members[?(@.value)]",
  "optional": true,
  "preserveArrayWithSingleElement": true,
  "functions": [
    {
      "function": "resolveEntityIds"
    }
  ]
}
]
}
}

```

5. Open the [Properties](#) tab and add the following properties and their values:

Field	Value
Authentication	BasicAuthentication
cflp.group.unique.attribute	externalId, ['urn:ietf:params:scim:schemas:extension:2.0:mapping'] ['providerId']
cflp.patch.group.members.above.threshold	5000
cflp.providerId	<ID of the content provider you created in your SAP Build Work Zone, standard edition tenant, e.g. S4_PC_UCT950>
cflp.user.unique.attribute	emails[0].value, ['urn:ietf:params:scim:schemas:extension:2.0:mapping'] ['providerId'],externalId
ips.trace.failed.entity.content	false

Field	Value
OAuth2TokenServiceURL	<Value of the UAA URL property of the SAP Build Work Zone, standard edition service key you created + /oauth/token>
Password	<Value of the clientsecret property of the SAP Build Work Zone, standard edition service key>
ProxyType	Internet
Type	HTTP
URL	<Value of the portal-service property of the SAP Build Work Zone, standard edition service key>
User	<Value of the clientid property of the SAP Build Work Zone, standard edition service key service key>

9. Navigate to the [Identity Provisioning](#) section and open the [Source Systems](#) section.
10. Select the source system you created for your SAP S/4HANA Cloud Private Edition tenant.
11. Open the [Jobs](#) tab.
12. Click [Run Now](#) for the [Read Job](#) to start the synchronization of your SAP S/4HANA Cloud Private Edition users and roles with your SAP Build Work Zone, standard edition tenant.
13. Navigate to the [Identity Provisioning](#) section and open the [Provisioning Logs](#) section to see the logs and status of your synchronization jobs.

You have now completed the setup of the Identity Provisioning service.

Users who have configured cards as visible in the Insights section of My Home in SAP S/4HANA Cloud, should now be able to see their cards in the [Insights](#) section of SAP Build Work Zone, standard edition.

Related Information

[Configure the Identity Provisioning Service](#)

4.5.5 Configure and Activate Joule Plug-In in a Target Mapping

Steps to configure and activate Joule plugin in SAP Fiori Launchpad.

You need to create a target mapping to activate the Joule plugin in the SAP Fiori Launchpad of your application. As a prerequisite, you must create a business catalog. For more information, see [Creating Catalogs](#).

Procedure

Procedure

1. In the launchpad app manager (see e.g. [Creating and Maintaining Launchpad App Descriptor Items](#)), create a target mapping with the intent `Shell-plugin` and the following configuration:

Field	Value
<i>Semantic Object</i>	Shell
<i>Action</i>	plugin
<i>Application Type</i>	SAP UI5 Fiori App
<i>Title</i>	Joule
<i>URL</i>	URL of the Joule application appended with <code>/resources/public/webclient/s4</code>
<i>ID</i>	sap.das.webclientplugin.s4
<i>Information</i>	Optional: Enter a description of your plug-in.
<i>Device Types</i>	Select the device or devices that are supported by your plug-in, e.g. desktop or phone. The plug-in will only be shown on these selected devices.
<i>Parameters</i>	Define parameters that you want to pass to the plug-in component by entering a name and a default value for each parameter.

Note

As target mappings with the intent `Shell-plugin` have a special meaning, they do not appear in standard services such as the `TargetMappings` feed, `resolveLink`, or `getLinksForSemanticObjects`.

2. In the launchpad content manager, assign the catalog to a role. See [Adding and Removing Catalogs, Groups and Spaces to/from Roles](#) for detailed information.
3. If you have not already done this, assign the role to the respective users (see [Assigning Users to Roles](#)).

4.5.6 Constraints for Joule in SAP S/4HANA Cloud Private Edition

In addition to the general [Constraints for Joule \[page 16\]](#), the following constraints apply for Joule in SAP S/4HANA Cloud Private Edition:

- Joule is available for SAP S/4HANA Cloud Private Edition 2023 FPS01 or later.
- A Joule instance can be only connected to exactly one SAP S/4HANA system.
- Joule is available for SAP S/4HANA Cloud Private Edition only within [Rise with SAP](#).
- You can enable Joule for SAP S/4HANA Cloud Private Edition only in an SAP managed data center setup.

4.6 Integration with SAP S/4HANA Cloud Public Edition

This section describes how to integrate Joule with the technical product SAP S/4HANA Cloud Public Edition. These steps comprise the Joule-specific and product-specific prerequisites and depend on your initial system setup. For example, you must first set up the technical environment, such as the SAP Business Technology Platform (BTP) with the entitlements for Joule and SAP Build Work Zone, standard edition (foundation/standard plan). You are guided through the integration steps with instructions, for example, you run the Joule booster that - among other settings - enables the communication scenario `SAP_COM_0882` (SAP Digital Assistant Services) in the background.

Note

SAP S/4HANA Cloud Public Edition refers to an offering that includes the technical product SAP S/4HANA Cloud Public Edition as the main component and a large variety of surrounding services. The integration described here is between Joule and the technical product SAP S/4HANA Cloud Public Edition.

Note

To access Joule within SAP S/4HANA Cloud Public Edition, an additional entitlement and authorization may be required. Please consult your account executive for more information.

Prerequisites

- You fulfill the general [Prerequisites \[page 7\]](#) for Joule.
- You have reviewed the [Initial Setup \[page 6\]](#) for Joule and have carried out the necessary steps.
- You checked the Joule-specific information regarding [Security](#).
- You are aware of the [Constraints for Joule in SAP S/4HANA Cloud Public Edition \[page 65\]](#).

→ Recommendation

If you have used SAP CoPilot Digital Assistant in the past, we recommend to request its deactivation (*CoPilot Operations* (**MOB-UIA-CP-OP**)) before setting up Joule. However, you can continue using SAP CoPilot Collaboration (human to human chat and collaboration capabilities of SAP CoPilot).

This recommendation is relevant for 2-system landscape only. For more information, see [System Landscapes in SAP S/4HANA Cloud](#).

Integration Steps

To integrate Joule with SAP S/4HANA Cloud Public Edition, you must carry out the following steps:

1. [Expose the SAP Fiori Launchpad Content to SAP BTP \[page 51\]](#) (specific for SAP S/4HANA Cloud Public Edition)
2. [Configure Trust to the Identity Authentication Tenant \[page 10\]](#)
3. [Configure User Attributes for Joule from the Identity Directory \[page 11\]](#)
4. [Register the SAP S/4HANA Cloud Public Edition System \[page 52\]](#) (specific for SAP S/4HANA Cloud Public Edition)
5. [Create SAP Build Work Zone Application and Instance \[page 20\]](#)
6. [Run the Joule Booster \[page 12\]](#)
7. [Configure Identity Provisioning Service \[page 53\]](#) (specific for SAP S/4HANA Cloud Public Edition)
8. [Configure Trusted Domain in SAP BTP \[page 16\]](#)
9. [Configure Trusted Domains in Identity Authentication \[page 15\]](#)
10. [Maintain Custom Content Security Policy \[page 65\]](#) (specific for SAP S/4HANA Cloud Public Edition)
11. [Set Up Document Grounding \[page 70\]](#)

Related Information

[Integrating With SAP Build Work Zone, Standard Edition](#)

4.6.1 Expose the SAP Fiori Launchpad Content to SAP BTP

To set up the content exposure flow, the Joule booster created a communication arrangement in the *Communication Arrangements* app, and a communication system for the communication scenario **SAP_COM_0647** (Launchpad Content Exposure to SAP BTP Integration). In most cases, you set the *Exposure Role Selection* to **ALL** (All Roles with SAP Fiori Launchpad Content). For more information, see [Expose Launchpad Content to SAP BTP](#).

However, if you would like to expose individual business roles to SAP BTP and have therefore set the [Exposure Role Selection](#) to **SELECTION** (Selected Roles in “Maintain Business Roles” Application), you must expose your roles as follows:

1. In the SAP S/4HANA Cloud Public Edition system, log on to the SAP Fiori Launchpad as administrator.
2. Open the [Maintain Business Roles](#) app.
3. Identify the business roles that you are using.
For more information about business roles in SAP S/4HANA Cloud Public Edition, see [Business Roles, Business Catalogs and Restrictions](#).
4. Open each role and select [Exposed to SAP BTP](#). For more information about mass changes, see [How to Select Business Roles for Exposure](#).
5. Confirm the dialog box that displays.
6. The [Exposed to SAP BTP](#) entry indicates that the role you selected is exposed.
7. The content is ready to be consumed by SAP Build Work Zone, standard edition. To look at the exposure log, open the application log in the [Display Launchpad Content Exposure Logs](#) tile.

4.6.2 Register the SAP S/4HANA Cloud Public Edition System

Procedure

1. In the SAP BTP cockpit, navigate to your global account, and then choose ► [System Landscape](#) ► [Systems](#) ►.
2. On the [Systems](#) page, choose [Add System](#). In the [Add System](#) wizard do the following:
 - a. Enter a name for the system you want to register.
 - b. In the [Type](#) dropdown list, select the system type.
 - c. Choose [Add](#).
 - d. Select [All Communication Scenarios](#) in the [Communication Scenario Groups](#) dropdown menu.
 - e. Choose [Get Token](#).

The system generates the registration token.
 - f. Copy the registration token. You need the token to complete the integration on the respective SAP solution system side.
 - g. Close the wizard.
3. Go to your SAP S/4HANA Cloud Public Edition system. Open the ► [Communication Management](#) ► [Maintain BTP Extension](#) ► app.
4. Choose [New](#) and add the token fetched from the SAP BTP cockpit as mentioned in the step 2.
5. The status of the system is set as [Enabled](#) once you add the token.
6. Go back to SAP BTP Cockpit and verify if the status of the system is set as [Registered](#).

4.6.3 Configure Identity Provisioning Service

Navigation service of SAP Build Work Zone, standard edition service uses Identity Provisioning service (IPS) that is used to provision identities and their authorizations between source and target systems.

This section describes the steps to configure the source system (SAP S/4HANA Cloud Public Edition) and target systems (Identity Authentication and SAP Build Work Zone, standard edition) in the Identity Provisioning of your IAS application user interface.

The following configurations are required to provision:

1. User details like user email, Global User ID and group memberships from SAP S/4HANA Cloud Public Edition to SAP Build Work Zone, standard edition.
2. User roles as groups from SAP S/4HANA Cloud Public Edition (source system) to SAP Build Work Zone, standard edition with role ID as external ID and group memberships.

Note

You must have the administrative access to your Identity Provisioning tenant and you need to enable the [Manage Identity Provisioning](#) role for your user. For more information, see [Manage Authorizations in SAP Cloud Identity Infrastructure](#).

You must have created the communication scenario and communication users for the communication arrangement SAP_COM_0193. For more information, see [SAP Cloud Identity Services](#).

Configure Source System

1. Login to your Identity Authentication tenant.
2. Navigate to ► [Identity Provisioning](#) ► [Source System](#) ►.
3. Add SAP S/4HANA Cloud Public Edition as a source system. For more information, see [Add New Systems](#).
4. Under the [Properties](#) tab, configure the connection settings for your system and add or confirm if the following properties are available.

Property Name	Value
Type	HTTP
URL	Enter the SAP S/4HANA Cloud Public Edition API URL. You can find the correct URL in the API-URL field of the communication arrangement set up for communication scenario SAP_COM_0193. For example: https://my123456-api.s4hana.ondemand.com or https://my123456-api.s4hana.cloud.sap.
ProxyType	Internet

Property Name	Value
Authentication	Enter your authentication method: BasicAuthentication
User	Enter the user name from the communication arrangement SAP_COM_0193 ⚠ Caution Do not use special symbol ',' (comma) as it is not supported.
Password	Enter the password for the user name from the communication arrangement SAP_COM_0193 ⚠ Caution Do not use special symbol ',' (comma) as it is not supported.
s4hana.cloud.skip.read.archived	In the event of archived (disabled) entities in the source system SAP S/4HANA Cloud Public Edition, choose whether the provisioning jobs to continue reading such entities or to skip them. This property is enabled by default. If you want to always read disabled entities, set the property to false , or delete it.
ips.date.variable.format	yyyy-MM-dd
s4hana.cloud.api.version	1 This means your SAP S/4HANA Cloud Public Edition system uses SAP_COM_0193 communication arrangement.
ips.trace.skipped.entity.content	true
ips.trace.failed.entity.content	true

- Configure the transformations. Transformations are used to map the user attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the SAP S/4HANA Cloud Public Edition source system, whose settings are displayed under the [Transformations](#) tab after saving its initial configuration. You can change the default transformation mapping rules depending on your setup of entities in your SAP S/4HANA Cloud Public Edition system.

Sample Code

```
{
  "user": {
    "condition": "($.user.validityPeriod.startDate <= '${currentDate}')
    && ($.user.validityPeriod.endDate > '${currentDate}')",
    "mappings": [
```

```

{
  "sourcePath": "$.personID",
  "targetVariable": "entityIdSourceSystem"
},
{
  "sourcePath": "$.personalInformation.firstName",
  "targetPath": "$.name.givenName",
  "optional": true
},
{
  "sourcePath": "$.personalInformation.lastName",
  "targetPath": "$.name.familyName",
  "optional": true
},
{
  "sourcePath": "$.personalInformation.middleName",
  "targetPath": "$.name.middleName",
  "optional": true
},
{
  "sourcePath": "$.personalInformation.personFullName",
  "targetPath": "$.name.formatted",
  "optional": true
},
{
  "sourcePath": "$.user.userName",
  "targetPath": "$.userName",
  "optional": true,
  "correlationAttribute": true
},
{
  "constant": true,
  "targetPath": "$.active"
},
{
  "condition": "$.user.lockedIndicator == 'true'",
  "constant": false,
  "targetPath": "$.active",
  "optional": true
},
{
  "sourcePath": "$.workplaceInformation.emailAddress",
  "targetPath": "$.emails[0].value",
  "optional": true,
  "correlationAttribute": true
},
{
  "sourcePath": "$.user.logonLanguageCode",
  "optional": true,
  "targetPath": "$.locale"
},
{
  "sourcePath": "$.PersonExternalID",
  "optional": true,
  "correlationAttribute": true
},
{
  "sourcePath": "$.user.globalUserID",
  "optional": true,
  "targetPath": "$.
[ 'urn:ietf:params:scim:schemas:extension:sap:2.0:User' ][ 'userUuid' ]"
},
{
  "sourcePath": "$.user.role",
  "targetPath": "$.groups",
  "preserveArrayWithSingleElement": true,
  "functions": [
    {

```

```

'null'",
    "condition": "'%s4hana.cloud.roles.prefix%' !=",
    "function": "concatString",
    "applyOnElements": true,
    "prefix": "%s4hana.cloud.roles.prefix%",
    "applyOnAttribute": "roleName",
    "assignToAttribute": "display"
  },
  {
    "condition": "'%s4hana.cloud.roles.prefix%' ==",
    "function": "concatString",
    "applyOnElements": true,
    "prefix": "",
    "applyOnAttribute": "roleName",
    "assignToAttribute": "display"
  },
  {
    "function": "concatString",
    "applyOnElements": true,
    "prefix": "",
    "applyOnAttribute": "roleName",
    "assignToAttribute": "value"
  }
]
},
{
  "type": "remove",
  "targetPath": "$.groups[*].roleName"
},
{
  "type": "valueMapping",
  "sourcePaths": [
    "$.user.timeZoneCode"
  ],
  "targetPath": "$.timezone",
  "defaultValue": "Europe/Berlin",
  "valueMappings": [
    {
      "key": [
        "WDFI"
      ],
      "mappedValue": "Europe/Berlin"
    },
    {
      "key": [
        "ISRAEL"
      ],
      "mappedValue": "Asia/Jerusalem"
    },
    {
      "key": [
        "RUS03"
      ],
      "mappedValue": "Europe/Moscow"
    },
    {
      "key": [
        "AUSNSW"
      ],
      "mappedValue": "Australia/Sydney"
    },
    {
      "key": [
        "UTC+4"
      ],
      "mappedValue": "Asia/Dubai"
    }
  ]
}

```



```

    {
      "key": [
        "BRAZIL"
      ],
      "mappedValue": "America/Sao_Paulo"
    },
    {
      "key": [
        "BRZLEA"
      ],
      "mappedValue": "America/Sao_Paulo"
    },
    {
      "key": [
        "MSTNO"
      ],
      "mappedValue": "America/Phoenix"
    },
    {
      "key": [
        "EST"
      ],
      "mappedValue": "America/New_York"
    },
    {
      "key": [
        "UTC"
      ],
      "mappedValue": "Etc/UTC"
    },
    {
      "key": [
        "UTC+3"
      ],
      "mappedValue": "Asia/Riyadh"
    },
    {
      "key": [
        "EST_"
      ],
      "mappedValue": "America/Toronto"
    },
    {
      "key": [
        "UTC+8"
      ],
      "mappedValue": "Asia/Shanghai"
    },
    {
      "key": [
        "JAPAN"
      ],
      "mappedValue": "Asia/Tokyo"
    }
  ]
},
{
  "type": "valueMapping",
  "sourcePaths": [
    "$.businessPartnerRoleCode"
  ],
  "targetPath": "$.userType",
  "defaultValue": "Employee",
  "valueMappings": [
    {
      "key": [
        "BUP003"
      ],

```

```

        "mappedValue": "Employee"
      },
      {
        "key": [
          "BBP005"
        ],
        "mappedValue": "Contingent Worker"
      }
    ]
  },
  "group": {
    "mappings": [
      {
        "sourcePath": "$.ID",
        "targetPath": "$.displayName",
        "targetVariable": "entityIdSourceSystem",
        "functions": [
          {
            "condition": "'%s4hana.cloud.roles.prefix%' !=",
            "function": "concatString",
            "prefix": "%s4hana.cloud.roles.prefix%"
          }
        ]
      },
      {
        "constant": "urn:ietf:params:scim:schemas:core:2.0:Group",
        "targetPath": "$.schemas[0]"
      },
      {
        "sourcePath": "$.to_BusinessUserAssignment.results",
        "optional": true,
        "preserveArrayWithSingleElement": true,
        "targetPath": "$.members"
      },
      {
        "type": "remove",
        "targetPath": "$.members[*].__metadata"
      },
      {
        "type": "remove",
        "targetPath": "$.members[*].UserName"
      },
      {
        "type": "rename",
        "constant": "value",
        "targetPath": "$.members[*].PersonID"
      },
      {
        "constant": "User",
        "targetPath": "$.members[*].type"
      }
    ]
  }
}

```

Configure Target System

1. Navigate to [Identity Provisioning > Target System](#).
2. Add [SAP Build Work Zone, standard edition](#) as a target system. For more information, see [Add New Systems](#).

3. Select SAP S/4HANA Cloud Public Edition as the [Source](#) system (that you configured in the previous step), whose identities you want to read and provision to the target one.
4. Choose the [Properties](#) tab to configure the connection settings for your system:

Property Name	Value
Type	HTTP
URL	Enter the API URL of SAP Build Work Zone, standard edition from the service key of your SAP Build Work Zone, standard edition instance under endpoints [portal-service]. It follows the pattern: https://portal-service.cfapps.sap.hana.ondemand.com
ProxyType	Internet
Authentication	BasicAuthentication
User	Enter the OAuth Client Id, from the service key of your SAP Build Work Zone, standard edition instance under uaa.clientid .
Password	(Credential) Enter the OAuth Client Secret, from the service key of your SAP Build Work Zone, standard edition instance under uaa.clientsecret .
OAuth2TokenServiceURL	Enter the OAuth 2.0 Token Service URL from the service key of your SAP Build Work Zone, standard edition instance. It follows the pattern: <uaa.url>/oauth/token . For example: https://ips-cflp-woaealle.authentication.sap.hana.ondemand.com/oauth/token
cflp.providerId	Enter a valid provider ID value. The provider ID is specified in the Channel Manager of the SAP Build Work Zone, standard edition when defining a new content provider.
ips.failed.request.retry.attempts	2
ips.failed.request.retry.attempts.interval	50
ips.trace.failed.entity.content	true
xsuaa.user.unique.attribute	userName,origin
xsuaa.origin	sap.custom

Property Name	Value
cfip.user.unique.attribute	If Identity Provisioning tries to provision a user that already exists in the target system (a conflicting user), this property defines the unique attributes by which the existing user will be searched and resolved. SAP Build Work Zone, standard edition supports the following unique attributes which are automatically filled in when the target system is added in the service UI: emails[0].value, ['urn:ietf:params:scim:schemas:extension:2.0:mapping'] ['providerId'],externalId • If the user has an externalId, the conflict is resolved by externalId and providerId. • If the user doesn't have an externalId, the conflict is resolved by email and providerId. For the conflict to be resolved, an existing user matching both unique attributes should be found. If an existing user doesn't match both unique attributes or matches only one of them, the user creation fails. → Tip We recommend that you do not modify the value of the cfip.user.unique.attribute property. Otherwise, user creation fails.

Property Name	Value
cflp.group.unique.attribute	If Identity Provisioning tries to provision a group that already exists in the target system (a conflicting group), this property defines the unique attributes by which the existing group will be searched and resolved. SAP Build Work Zone, standard edition supports a pair of unique attributes which is automatically filled in when the target system is added in the service UI: externalId, ['urn:ietf:params:scim:schemas:extension:2.0:mapping']['providerId'] For the conflict to be resolved, an existing group matching both unique attributes should be found. In this case, Identity Provisioning updates the group. This means, the conflicting group overwrites the existing one. If the group matches only one of the unique attributes, the conflict is not resolved, and the group creation fails. → Tip We recommend that you do not modify the value of the cflp.user.unique.attribute property. Otherwise, user creation fails.

Property Name	Value
cfip.support.bulk.operation	This property enables bulk operations for users and groups. When the property is enabled (set to <i>true</i>), the following operations can be executed by Identity Provisioning service in a single request: • Create or delete multiple users • Create, update, or delete multiple groups When the property is disabled (set to <i>false</i>), the following operations can be executed by Identity Provisioning service for a single entity at a time: • Create or delete a user • Create, update, or delete a group Note Update operation is skipped for users in the default write transformation. Possible values: • true • false Default value: false
cfip.bulk.operations.max.count	If you have enabled the bulk operations, you can use this property to set the number of operations to be provisioned per request. Possible values: Default value: 20 Maximum value: 100 If you enter a number larger than 100, the service will replace it with the default value (20).
(Optional) ips.delete.threshold.groups	Use this property to control the number of groups to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of groups, for example by adding a filter or condition. For more information, see List of Properties
(Optional) ips.delete.threshold.users	Use this property to control the number of users to be deleted in a target system by defining a threshold. This will prevent you from accidentally deleting a huge number of users, for example by adding a filter or condition. For more information, see List of Properties

- From the left panel and go to the [Transformations](#) tab. You can switch to the JSON editor by choosing the code-bracket icon. Transformations are used to map user and group attributes from the data model of the source system to the data model of the target system, and the other way around. The Identity Provisioning offers a default transformation for the SAP Build Work Zone, standard edition target system, whose settings are displayed under the [Transformations](#) tab after saving its initial configuration.

Sample Code

```
{
  "user": {
    "skipOperations": [
      "update"
    ],
    "mappings": [
      {
        "sourceVariable": "entityIdTargetSystem",
        "targetPath": "$.id"
      },
      {
        "constant": "urn:ietf:params:scim:schemas:core:2.0:User",
        "targetPath": "$.schemas[0]"
      },
      {
        "constant": "%cflp.providerId%",
        "targetPath": "$[urn:ietf:params:scim:schemas:extension:2.0:mapping]['providerId']"
      },
      {
        "condition": "$.emails[?(@.primary == true)].value == []",
        "sourcePath": "$.emails[0].value",
        "optional": true,
        "targetPath": "$.emails[0].value"
      },
      {
        "condition": "$.emails[?(@.primary == true)].value != []",
        "sourcePath": "$.emails[?(@.primary == true)].value",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.emails[0].value",
        "functions": [
          {
            "function": "elementAt",
            "index": 0
          }
        ]
      },
      {
        "condition": "$.emails[0].length() > 0",
        "constant": true,
        "targetPath": "$.emails[0].primary"
      },
      {
        "sourcePath": "$[urn:ietf:params:scim:schemas:extension:sap:2.0:User]['userUuid']",
        "optional": true,
        "targetPath": "$.externalId"
      },
      {
        "sourcePath": "$.groups[*].value",
        "preserveArrayWithSingleElement": true,
        "optional": true,
        "targetPath": "$.groups[?(@.value)]",
        "functions": [
          {
            "entityType": "group",
```

```

        "function": "resolveEntityIds"
      }
    ]
  }
],
"group": {
  "mappings": [
    {
      "sourceVariable": "entityIdTargetSystem",
      "targetPath": "$.id"
    },
    {
      "constant": "%cflp.providerId%",
      "targetPath": "$
[ 'urn:ietf:params:scim:schemas:extension:2.0:mapping' ][ 'providerId' ]"
    },
    {
      "constant": "urn:ietf:params:scim:schemas:core:2.0:Group",
      "targetPath": "$.schemas[0]"
    },
    {
      "constant": "urn:ietf:params:scim:schemas:core:2.0:mapping",
      "optional": true,
      "targetPath": "$.schemas[1]"
    },
    {
      "sourcePath": "$.displayName",
      "optional": true,
      "targetPath": "$.externalId"
    },
    {
      "sourcePath": "$.externalId",
      "optional": true,
      "targetPath": "$.externalId",
      "functions": [
        {
          "function": "replaceAllString",
          "regex": "(?i)(^pcd:)",
          "replacement": ""
        },
        {
          "function": "replaceString",
          "target": "/",
          "replacement": ":"
        },
        {
          "function": "replaceString",
          "target": "(",
          "replacement": "@"
        },
        {
          "function": "replaceString",
          "target": ")",
          "replacement": "+"
        }
      ]
    }
  ],
  {
    "sourcePath": "$.members[*].value",
    "preserveArrayWithSingleElement": true,
    "optional": true,
    "targetPath": "$.members[?(@.value)]",
    "functions": [
      {
        "function": "resolveEntityIds"
      }
    ]
  }
]

```



```
}  
  }  
    ]  
      }
```

Run Replication in IPS

In order to replicate content from SAP S/4HANA Cloud Public Edition system to the content provider in SAP Build Work Zone standard edition, you must run the provisioning job from the Identity Provisioning user interface (UI).

1. Under [Source Systems](#), select your source system for SAP S/4HANA Cloud Public Edition. Choose ► [Jobs](#) ► [Read Job](#) ► [Run Now](#) ►.
2. Choose [Resync](#) and choose [Run Now](#).

Note

It is recommended to schedule the [Read Job](#) once every twenty four hours.

4.6.4 Maintain Custom Content Security Policy

Maintain a custom content security policy to enable loading the plugin script as cross-origin resource.

1. Open [Manage Content Security Policy](#) app, click [Trusted Sites](#) and navigate to details for Resources Scripts.
2. Click [New](#) in [Managed by Customer](#) section.
3. Enter the URL of your Joule tenant under [Trusted Content](#) and choose [Save](#).

4.6.5 Constraints for Joule in SAP S/4HANA Cloud Public Edition

In addition to the general [Constraints for Joule \[page 16\]](#), the following constraints apply for Joule in SAP S/4HANA Cloud Public Edition:

- Joule does not cover the full set of globalization support as SAP S/4HANA Cloud Public Edition. For now, you can converse with Joule in English (EN).
- Joule is not designed to process personal data or sensitive data. You must not add any personal sensitive data in Joule. For more information, see [Data Protection and Privacy](#).
- Joule is designed to work with one SAP S/4HANA Cloud Public Edition instance only.
- To use single log-out (SLO) for Joule with SAP S/4HANA Cloud Public Edition, you need to configured (SLO). For more information, see [Authentication of Requests](#).
- In the Joule booster, when you enable Joule in the system, Joule is enabled for all users.

Since Joule Web Client is embedded via iFrame, it requires support for iFrame and 3rd-party cookies. This implies the following technical constraints:

- Blocking of iFrame from Joule domains (* .cloud.sap) and IAS domains must be disabled in the browser settings, either by default or by policy. See [Supported Browsers](#) for details.
- Single Sign-On (SSO) relies on an a shared session with IAS used by SAP S/4HANA Cloud Public Edition instance and Joule instance. Thus, IAS must not be configured to force authentication with the Corporate IdP for each authentication if IAS is configured as a proxy IdP.

Other technical constraints:

- Rate limits for BTP apply.
- Rate limit for Joule is 3000 requests per minute per tenant.
- Restrictions for SAP Build Work Zone, standard edition with SAP S/4HANA Cloud Public Edition apply. For more information, see [3075638](#) .
- If SAP Build Work Zone, standard edition is not already used, Joule uses SAP Build Work Zone, standard edition with foundation plan for navigational scenarios. This implies that Identity Provisioning Service (IPS) is enabled and will be configured for user assignment to the navigation content.

4.7 Integration with SAP Integrated Product Development

SAP Integrated Product Development is a cloud solution that enables you to digitally orchestrate your product development from design to operate, resulting in reduced time to market, higher R&D return on investment, higher margins, and reliable product launches.

With Joule, business users have conversational access to the SAP Integrated Product Development product documentation on the SAP Help Portal from within the SAP Fiori apps that are delivered for the different SaaS applications. For more information about supported scope and setting up Joule for SAP Integrated Product Development, see [3503056](#) .

4.8 Integration with SAP Integrated Business Planning for Supply Chain

SAP Integrated Business Planning for Supply Chain (SAP IBP) combines supply chain monitoring, sales and operations planning, demand management, inventory planning, and supply planning into a comprehensive solution.

With Joule, business users have conversational access to the SAP IBP product documentation on SAP Help Portal from the Fiori launchpad of their SAP IBP system.

Note

- To access Joule within SAP IBP, an additional entitlement and authorization may be required. Please consult your account executive for more information.
- Region restrictions may apply.
- For details about supported scope and setting up Joule for SAP IBP, see SAP Note [3497247](#) .

Prerequisites

- You've read and understood the relevant security information for Joule. For more information, see [Security](#).
- You're using the same email address for your SAP IBP system and in SAP Cloud Identity Services - Identity Authentication.

Prerequisites related to SAP Business Technology Platform (SAP BTP)

Joule is an SAP BTP based solution, so you need working knowledge in the following topics: entitlements, subaccount, system landscape, booster, and you need to set up SAP BTP. This includes the following requirements:

- You have an enterprise global account on SAP BTP with license type CUSTOMER and have configured your global account with the required entitlements.
 - In addition, you have registered your SAP IBP system in your global account. For more information, see [Register an SAP Integrated Business Planning System in a Global Account in SAP BTP](#) and [Trigger the Registration in the SAP Integrated Business Planning Tenant](#).
- You've created a subaccount for your SAP IBP system in your SAP BTP global account. In addition, you've completed the following configuration steps for your subaccount:
 - You've established trust between SAP Authorization and Trust Management Service and SAP Cloud Identity Services - Identity Authentication in your subaccount. For more information, see [Configure Trust to the Identity Authentication Tenant \[page 10\]](#).
 - You've configured the `user_uuid` assertion attribute to the `Global User ID` field in the Identity Authentication application corresponding to your subaccount to allow user identification based on global user ID. For more information, see [Global User ID in Integration Scenarios](#). For a step-by-step procedure, see [Configure User Attributes for Joule from the Identity Directory \[page 11\]](#).
 - You've added the entitlement for the connectivity plan of the SAP Cloud Identity services in your subaccount.

Once you've completed the SAP BTP prerequisites, check whether you fulfill the general prerequisites for Joule. For more information, see [Prerequisites \[page 7\]](#).

Integration Process

When you fulfill all prerequisites, you can start the actual integration. It consists of the following steps:

1. You need to create a formation in the SAP BTP global account that includes your Joule system. To do this, run the [Setting up Joule](#) booster. This booster creates a formation including Joule and the SAP IBP system that you selected for the integration. For more information, see [Run the Joule Booster \[page 12\]](#).

Note

The [Setting up Joule](#) booster creates a communication arrangement and communication system in your SAP IBP system for the [SAP Digital Assistant Services \(SAP_COM_0882\)](#) communication scenario. Do not create, change or remove this communication arrangement and system from your SAP IBP system. Otherwise, the Joule integration will fail.

2. Once the Joule formation is created in your system, you must maintain the custom content security policy in your SAP IBP tenant to enable loading the Joule icon. For more information, see [Maintaining Custom Content Security Policy \[page 68\]](#).

Verifying the Integration

After you've completed the integration process, Joule should be available in your SAP IBP system. To verify this, navigate to the SAP Fiori launchpad of your SAP IBP system and check whether the Joule icon  appears in the upper right part of your shell bar.

4.8.1 Maintaining Custom Content Security Policy

You need to add the Joule tenant URL as a trusted source to enable loading Joule in your SAP Integrated Business Planning for Supply Chain (SAP IBP) system.

Context

Once you've run the booster, you must maintain the custom content security policy in your SAP IBP tenant to enable loading the Joule icon .

Procedure

1. Navigate to the SAP Fiori launchpad of your SAP IBP system.
2. Open the *Manage Content Security Policy* app.
3. On the *Trusted Sites* tab, open the details for *Resources Scripts* (UI_RESOURCES_SCRIPTS).
4. Under *Managed by Customer*, choose *New*.

→ Tip

If you don't see the *Managed by Customer* section, choose *Add*, and the section is added.

5. Under *Trusted Content*, enter the URL of your Joule tenant and choose *Save*.

ⓘ Note

To find out the URL of your Joule tenant, go to your SAP Business Technology Platform (SAP BTP) subaccount and navigate to ► *System Landscape* ► *Formations* ►. Select the formation that was created by the Joule booster and view its details. Copy the URL and enter it as a trusted site in your SAP IBP system.

For example: **`https://<subaccount>.<region>.sapdas.cloud.sap`**

4.8.2 Constraints for Joule in SAP Integrated Business Planning for Supply Chain

Familiarize yourself with the constraints for Joule in SAP Integrated Business Planning for Supply Chain (SAP IBP).

For more information about constraints for Joule, see [Constraints for Joule \[page 16\]](#).

In addition, the following constraints apply:

- Joule doesn't support the same set of accessibility standards as SAP IBP.
- Joule is designed to work with one SAP IBP instance only. So if, for example, you want to integrate Joule with the SAP IBP instance of your test system and the SAP IBP instance of your productive system, you need separate Joule tenants for each of the systems. Equally, if you have more than one productive system and want to integrate Joule with each of them, you need separate Joule tenants for each of the systems.
- Custom domains aren't supported.
- Joule isn't designed to process personal data or sensitive data. You must not add any personal or sensitive data in Joule. For more information, see [Data Protection and Privacy](#).
- The Joule Web client is embedded via inline frame (iFrame), so it requires support for iFrame and third-party cookies. This implies the following technical constraints:
 - Blocking of iFrame from Joule domains (*.cloud.sap) and SAP Cloud Identity Services - Identity Authentication (IAS) domains must be disabled in the browser settings, either by default or by policy. For more information, see [Supported Browsers](#).

Note

Joule supports cookie partitioning to mitigate the impact of the deprecation of third-party cookies by Google Chrome. Users can continue to leverage single sign-on-based login to Joule after the third-party cookies are deprecated.

- Single sign-on relies on a shared session with the IAS used by the SAP IBP instance and the Joule tenant. If the IAS is configured to use a proxy IdP, the IAS mustn't be configured to force authentication with the corporate IdP for each authentication.
- In the [Setting up Joule](#) booster, when you enable Joule in your SAP IBP system, it is enabled for all users.

5 Set Up Document Grounding

The document grounding capability uses information gathered from documents that you upload to the service to add context to Joule answers. To make this optional capability available, first complete the integration steps between Joule and the SAP solution that you want to use to consume Joule. Then, set up document grounding as described here.

For now, the scope of the document grounding capability is as follows:

- Data repository: Microsoft SharePoint (You can upload a maximum of 2000 documents to the service.)
- Document (file) format:
 - HTML (Static)
 - JPEG
 - JPG
 - Microsoft Word
 - PDF
 - PNG
 - TIFF
- Document content: Plain text and images (Tables and images aren't currently supported.)
- Content refresh: SharePoint content updated once every day

Before triggering the configuration process, make sure that you've done the following for your global account and subaccount:

- You've purchased the AI Unit SKU (8018592) and AI units. After you've done so, the entitlements for document grounding are automatically added to your global account. For more information, contact your account executive.
- You've added the entitlements for document grounding to the subaccount where you've subscribed to Joule. For more information, see [Managing Entitlements and Quotas Using the Cockpit](#).
- You've enabled Identity Authentication. For more information, see [What Are Cloud Identity Services?](#).

⚠ Caution

The document grounding capability has no mechanism for determining the type of data that it processes. The capability therefore can't determine whether this data includes personal data, which is subject to the data protection laws applicable in specific countries/regions.

Any ingested documents are available to all Joule business users in the given Joule tenant. We therefore recommend that you don't expose confidential or other privileged documents.

📌 Note

For now, the response from Joule doesn't take context information into consideration, for example, user location and user permission.

The configuration process involves the following steps:

- [Configure User Authentication \[page 71\]](#)

- [Prepare SharePoint Integration \[page 74\]](#)
- [Configure Access from SAP BTP \[page 74\]](#)
- [Set Up Content Ingestion \[page 77\]](#)

See also the blog post [Getting Started with Document Grounding for Your SAP SFSF](#).

Related Information

[Metering and Pricing \[page 78\]](#)

5.1 Configure User Authentication

Configure user authentication and access the document grounding capability.

To access the document grounding capability, complete the following configuration steps:

1. Open the [SAP BTP cockpit](#) and go to the subaccount where you've subscribed to Joule.
2. Navigate to [Services > Service Marketplace](#).
3. Search for **document grounding** and choose the tile.
4. Choose [Create](#).
5. Choose **Other** as *Runtime Environment*. Enter an *Instance Name*, copy it, and paste it into a text editor. You need the instance name in step 12.
Choose [Create](#).
6. Choose [View Instance](#).
7. Select the instance and create a service binding. Enter a *Service Binding Name*.
8. Choose [Create](#). Once the service binding is created, choose its name to display the details. Copy the service binding URL and paste it into a text editor. You need this URL in step 21.
9. Again, navigate to [Services > Service Marketplace](#).
10. Search for **Cloud Identity Services** and choose the tile.
11. Choose [Create](#).
Choose **application** as *Plan* and **Other** as *Runtime Environment*. Enter an *Instance Name*.
Choose [Next](#).
12. In *Parameters*, enter the following parameter, using the *Instance Name* that you created for document grounding in step 5 as the "service-instance-name" value:

```
{
  "consumed-services": [
    {
      "service-instance-name": "<doc-grounding-instance-name>"
    }
  ]
}
```

13. Choose [Create](#).

14. Choose [View Instance](#).
15. Select the instance and create a service binding. Enter a [Service Binding Name](#) and the following parameters:

```
{
  "credential-type": "X509_GENERATED",
  "validity": 365,
  "validity-type": "DAYS"
}
```

→ Remember

SAP Cloud Identity Services triggers an automatic e-mail when the certificate is nearing its expiration date. To generate the new certificate, re-create the service binding.

16. Choose [Create](#).
17. Once the service binding is created, choose its name to display the details.
18. Copy the values for "clientid" and "authorization_endpoint" from the service binding and paste them into a text editor.

📘 Note

Edit the "authorization_endpoint" value by replacing the ending .../oauth2/authorize with .../oauth2/token.

You need the values for "clientid" and "authorization_endpoint" in step 20.

19. Create two files (for example, doc-grounding.crt and doc-grounding.key) using the service binding values for "certificate" and "key". The files can have any name.

📘 Note

The service binding values are generated with the characters "\n" representing new lines, instead of actual line breaks. To use the doc-grounding.crt and doc-grounding.key files, convert the "\n" characters into real line breaks. You can do so either manually, replacing each "\n" with a line break, or using a bash function, as in the following examples:

For MacOS:

```
sed 's/\\n/\\n/g' doc-grounding.crt > temp.crt && mv temp.crt doc-grounding.crt
```

```
sed 's/\\n/\\n/g' doc-grounding.key > temp.key && mv temp.key doc-grounding.key
```

For Windows PowerShell:

```
(Get-Content -Path doc-grounding.crt -Raw) -replace '\\n', "`n" |
Set-Content -Path temp.crt; Move-Item -Path temp.crt -Destination doc-grounding.crt -Force
```

```
(Get-Content -Path doc-grounding.key -Raw) -replace '\\n', "`n" |
Set-Content -Path temp.key; Move-Item -Path temp.key -Destination doc-grounding.key -Force
```


20. To get a token, submit the following request in the same folder you downloaded the `doc-grounding.crt` and `doc-grounding.key` files:

```
curl \
--request POST \
--url <edited_authorization_endpoint> \
--header 'accept: application/json' \
--header 'content-type: application/x-www-form-urlencoded' \
--data 'client_id=<clientid>' \
--data 'grant_type=client_credentials' \
--cert <file_with_certificate> \
--key <file_with_key>
```

Placeholder	Description
<edited_authorization_endpoint>	The "authorization_endpoint" value that you obtained in step 18, edited as described in step 18
<clientid>	The "clientid" value that you obtained in step 18
<file_with_certificate>	The <code>doc-grounding.crt</code> file, edited with line breaks as described in step 19
<file_with_key>	The <code>doc-grounding.key</code> file, edited with line breaks as described in step 19

You should receive a response like the following:

```
{
  "access_token":"eyJq.....LI-L8Ks0QV593dmtPU1g",
  "token_type":"Bearer",
  "expires_in":3600
}
```

Note

You need the token to [Set Up Content Ingestion \[page 77\]](#).

21. Use the service binding URL, the token, and the `doc-grounding.crt` and `doc-grounding.key` files to call the document grounding endpoints:

```
curl \
--request GET \
--url '<url>/pipeline/api/v1/pipeline' \
--header 'accept: application/json' \
--header 'Authorization: Bearer <access_token>' \
--cert <file_with_certificate> \
--key <file_with_key>
```

Placeholder	Description
<url>	The service binding URL value for document grounding that you obtained in step 8
<file_with_certificate>	The <code>doc-grounding.crt</code> file, edited with line breaks as described in step 19

Placeholder	Description
<file_with_key>	The <code>doc-grounding.key</code> file, edited with line breaks as described in step 19

You should receive a response with an empty list [] since the pipeline hasn't been created yet.

5.2 Prepare SharePoint Integration

Create user and register application in Microsoft Entra ID.

1. Create a technical user. For more information, see the Microsoft Entra documentation article [Quickstart: Create and assign a user account](#) .
2. Register an application. For more information, see the Microsoft Entra documentation article [Quickstart: Register an application with the Microsoft identity platform](#) .

Note

In *API permissions*:

- If you configure access from SAP BTP with **OAuth2Password** authentication, use the permission name **Sites.Read.All** with type **Delegated**
- If you configure access from SAP BTP with **OAuth2ClientCredentials** authentication, use the permission name **Sites.Selected** for Graph with type **Application**

For more information, see [Configure Access from SAP BTP \[page 74\]](#).

3. Add a scope to the application that you've registered. For more information, see the Microsoft Entra documentation article [Quickstart: Configure an application to expose a web API](#) .

5.3 Configure Access from SAP BTP

Create a destination configuration in the SAP BTP cockpit using either **OAuth2Password** or **OAuth2ClientCredentials** authentication.

With **OAuth2Password** *Authentication*

1. Open the [SAP BTP cockpit](#).

- Go to the subaccount where you've subscribed to Joule, choose **Connectivity** > **Destinations**, and create a new destination with the following information:

Field	Value
<i>Name</i>	<NAME_OF_DESTINATION>
<i>Type</i>	HTTP
<i>URL</i>	https://graph.microsoft.com
<i>Proxy Type</i>	Internet
<i>Authentication</i>	OAuth2Password
<i>User</i>	The technical user that you've created in Microsoft Entra ID
<i>Password</i>	The password that you've created in Microsoft Entra ID
<i>Client ID</i>	Microsoft Entra ID Application credentials
<i>Client Secret</i>	Microsoft Entra ID Application credentials
<i>Token Service URL</i>	https://login.microsoft.com/<TENANT_ID>/oauth2/v2.0/token , where <TENANT_ID> is the tenant ID of the Microsoft Entra ID Application

- In the *Additional Properties* section, enter the following information:

Field	Value
<i>scope</i>	https://graph.microsoft.com/.default

- Save your entries.
- Choose *Check Connection* to check whether the basic configuration is working.

Destination Configuration Sample (OAuth2Password authentication)

```
Name=<NAME_OF_DESTINATION>
Description=SharePoint Access (On-behalf of technical user)
Type=HTTP
ProxyType=Internet
Authentication=OAuth2Password
clientId=<CLIENT_ID>
clientSecret=<CLIENT_SECRET>
User=<USERNAME>
Password=<PASSWORD>
tokenServiceURL=https://login.microsoft.com/<TENANT_ID>/oauth2/v2.0/token
tokenServiceURLType=Dedicated
scope=https://graph.microsoft.com/.default
URL=https://graph.microsoft.com
```

With **OAuth2ClientCredentials** Authentication

1. Open the [SAP BTP cockpit](#).
2. Go to the subaccount where you've subscribed to Joule, choose [Connectivity](#) > [Destinations](#), and create a new destination with the following information:

Field	Value
Name	<NAME_OF_DESTINATION>
Type	HTTP
URL	https://<SHAREPOINT_DOMAIN>/sites/<NAME_OF_SHAREPOINT_SITE>
Proxy Type	Internet
Authentication	OAuth2ClientCredentials
Client ID	Microsoft Entra ID Application credentials
Client Secret	Microsoft Entra ID Application credentials
Token Service URL	https://login.microsoft.com/<TENANT_ID>/oauth2/v2.0/token, where <TENANT_ID> is the tenant ID of the Microsoft Entra ID Application

3. In the [Additional Properties](#) section, enter the following information:

Field	Value
scope	https://graph.microsoft.com/.default

4. Save your entries.
5. Choose [Check Connection](#) to check whether the basic configuration is working.

Destination Configuration Sample (**OAuth2ClientCredentials** authentication)

```
Name=<NAME_OF_DESTINATION>
Description=SharePoint Access (On-behalf of technical user)
Type=HTTP
ProxyType=Internet
Authentication=OAuth2ClientCredentials
clientId=<CLIENT_ID>
clientSecret=<CLIENT_SECRET>
tokenServiceURL=https\://login.microsoft.com/<TENANT_ID>/oauth2/v2.0/token
tokenServiceURLType=Dedicated
scope=https\://graph.microsoft.com/.default
URL=https\://<SHAREPOINT_DOMAIN>/sites/<NAME_OF_SHAREPOINT_SITE>
```

5.4 Set Up Content Ingestion

Configure schedule for scanning SharePoint and ingesting documents.

To register a new data pipeline, submit a POST request to the endpoint `<url>/pipeline/api/v1/pipeline` with the desired configuration parameters.

→ Tip

The `<url>` value is the service key URL value for document grounding that you obtained in step 8 in [Configure User Authentication \[page 71\]](#).

Request Example

```
curl \
--request POST \
--url '<url>/pipeline/api/v1/pipeline' \
--header 'Authorization: Bearer <access_token>' \
--header 'content-type: application/json' \
--data '{"type": "MSSharePoint", "configuration":
{"destination": "<NAME_OF_DESTINATION>", "sharePoint":
{"site": {"name": "<NAME_OF_SHAREPOINT_SITE>", "includePaths": ["/
<NAME_OF_SHAREPOINT_FOLDER_1>", "/<NAME_OF_SHAREPOINT_FOLDER_2>"]}}}' \
--cert <file_with_certificate> \
--key <file_with_key>
```

📌 Note

In the request, the parameter `includePaths` is optional. Use it if you want to add specific SharePoint folders from the SharePoint site to the data pipeline.

Nested folders are supported. Use `"includePaths": ["/<NAME_OF_SHAREPOINT_FOLDER_1>/<NAME_OF_SHAREPOINT_FOLDER_2>"]` if `<NAME_OF_SHAREPOINT_FOLDER_2>` is a subfolder of `<NAME_OF_SHAREPOINT_FOLDER_1>`.

Response Example

The response is given as a JSON file that contains the ID of the created pipeline.

```
{
  "pipelineId": "4cfd0478-29ea-45c2-bc40-d3817621744e"
}
```

The created pipeline starts ingesting the data in the provided SharePoint site and updates the content daily.

5.5 Metering and Pricing

Find out about metering and pricing for the document grounding capability.

→ Tip

The metering and pricing details presented here are relevant only to users of the **data-manager** service plan for enterprise accounts.

Usage Metric

The document grounding capability is metered monthly based on the predefined usage metric – records.

One record corresponds to one 50-MB block of data uploaded to the cloud service. For pricing purposes, one record is equivalent to 0.005 AI units.

The following table shows examples of how the size of an uploaded document determines the number of records:

Document Size	Number of Records
45 MB	1
80 MB	2
105 MB	3

Price Calculation

The total monthly price for the document grounding capability is calculated as follows:

- Number of records for the month * 0.005 * price per AI unit

ⓘ Note

The number of records for the month is the total number of records added to the cloud service during that month plus any records carried forward from the preceding month. Records remaining in the cloud service at the end of one month are automatically carried forward to the next.

If you delete uploaded documents, they're removed from the cloud service. However, they're not deducted from the total number of chargeable records for the month in which you delete them. Deleted documents aren't carried forward to the following month.

Examples

Key Elements of the Monthly Price Calculation

The following table illustrates how the number, size, and deletion of documents affect the number of records used in the monthly price calculation.

Number of Documents Uploaded in Month	Size Per Document	Number of Documents Deleted	Number of Records	Calculation of Monthly Price
1	50 MB	0	1	$1 * 0.005 * \text{price per AI unit}$
1	52 MB	0	2	$2 * 0.005 * \text{price per AI unit}$
10	5 MB	0	10	$10 * 0.005 * \text{price per AI unit}$
25	50 MB	10	25	$25 * 0.005 * \text{price per AI unit}$
25	51 MB	10	50	$50 * 0.005 * \text{price per AI unit}$

Monthly Price Calculation Including Records from Earlier Months

Let's say you start using the cloud service in June. At the outset, there are no documents in the service. Over the course of the month, you upload documents equivalent to a total of 10 records. Before the end of June, you delete documents equivalent to 5 records, leaving 5 records in the cloud service at the end of the month.

In this case, the monthly charge for June is calculated as follows:

- Total number of records added to cloud service in current month (10) * 0.005 * price per AI unit

→ Remember

Deleting documents reduces the number of records in the cloud service. But it doesn't change the total number of records for the current month.

The 5 records in the cloud service at the end of June are carried forward to the following month. These records now count toward the total number of records for July.

In July, you again upload documents equivalent to a total of 10 records and delete the equivalent of 5.

The monthly charge for July is calculated as follows:

- (Number of records carried over from previous month (5) + total number of records added to cloud service in current month (10) = 15) * 0.005 * price per AI unit

The number of records remaining at the end of July is calculated as follows:

- (Number of records carried over from previous month (5) + total number of records added to cloud service in current month (10) = 15) – number of records deleted during month (5) = 10

These 10 records are carried forward to August.

This process applies for all subsequent months.

→ Tip

Use the [SAP Discovery Center - AI Feature Estimator](#) .

Related Information

[SAP AI Services List](#) 

6 Troubleshooting

This section contains information that helps you troubleshoot problems in Joule.

Joule screen is blank

Root Cause

You are using Corporate Identity Provider (like Azure AD or OKTA) for user login.

Solution

- Make sure you have set your Corporate Identity Provider as the Default Identity Provider for Joule application in the Identity Authentication.
For more information, see [Choose Default Identity Provider for an Application](#).

Joule Navigation button is missing after a successful Jobrun or Jobsync

Solution

- Check your NavigationService destination settings in your SAP BTP subaccount.
- Open SAP Build Work Zone application in your SAP BTP subaccount and select Content Channel that you have created for your product integration.
- Ensure to select the *Token Service URL Type* as *Dedicated*.
- Manually refresh SAP Build Work Zone application to check if service provider details are updated.

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